

How Animals Move: Comparative Lessons on Animal Locomotion

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ABSTRACT

Comparative physiology often provides unique insights in animal structure and function. It is specifically through this lens that we discuss the fundamental properties of skeletal muscle and animal locomotion, incorporating variation in body size and evolved difference among species. For example, muscle frequencies *in vivo* are highly constrained by body size, which apparently tunes muscle use to maximize recovery of elastic recoil potential energy. Secondary to this constraint, there is an expected linking of skeletal muscle structural and functional properties. Muscle is relatively simple structurally, but by changing proportions of the few muscle components, a diverse range of functional outputs is possible. Thus, there is a consistent and predictable relation between muscle function and myocyte composition that illuminates animal locomotion. When animals move, the mechanical properties of muscle diverge from the static textbook force-velocity relations described by A. V. Hill, as recovery of elastic potential energy together with force and power enhancement with activation during stretch combine to modulate performance. These relations are best understood through the tool of work loops. Also, when animals move, locomotion is often conveniently categorized energetically. Burst locomotion is typified by high-power outputs and short durations while sustained, cyclic, locomotion engages a smaller fraction of the muscle tissue, yielding lower force and power. However, closer examination reveals that rather than a dichotomy, energetics of locomotion is a continuum. There is a remarkably predictable relationship between duration of activity and peak sustainable performance. © 2013 American Physiological Society. *Compr Physiol* 3:289-314, 2013.

Introduction

“Divide each difficulty into as many parts as is feasible and necessary to resolve it.”

—Rene Descartes

“One can’t see the forest for the trees.”

—Proverb

One biological premise not debated is that the ability to move about has been a persistent theme in animal evolution. The consequence of strong and unrelenting selection is that extant animals employ a diversity of structural and functional characteristics to move through air; through, on and under water; and across virtually all terrestrial habitats. The motor that powers animal movement, skeletal muscle, is both conservative and relatively simple in design. Within the vertebrates, the skeletal elements that translate muscle force into movement are also conserved across taxa, just as is the muscle exoskeleton within the arthropods. Thus, animal locomotion is a story of adapting a basic motor-skeletal system to serve various tasks. This article is based on the premise that conserved design results in a suite of “transcendent” properties that cross-taxa and even modes of transport while recognizing that the specific demands of swimming, flying, and running have resulted in unique solutions. Thus, this review will focus primarily on the unifying patterns evident among animals. Are there apparent “rules” constraining design and function?

The comparative process

Of course, any review built on a comparative physiology approach should pay appropriate homage to August Krogh, the oft-cited father of Comparative Physiology. While the “Krogh Principle” (120) is usually interpreted to focus on a specific species ideally suited for the study of a particular physiological problem, we take a broader view of Krogh’s comparative physiology principle here. Only by looking simultaneously at multiple species do patterns emerge which suggest principles of design. These transcendent principles provide unique insights into animal design that single species studies never could. Thus, while we also recognize Descartes, we emphasize that the reductionist paradigm inspired by his writing, successful though it has been, is incapable of identifying the larger scale patterns described below. Rather, seeking inspiration in the comparative approach, we describe unifying principles (the forest) that disappear in isolation.

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A year before Krogh's paper appeared in *Science*, J.B.S. Haldane wrote the compelling essay "On Being the Right Size" (76). In this engaging essay, he makes a strong case that body size is the most obvious distinguishing feature among animals, one that dictates virtually all aspects of form and function. Body size is itself a natural experiment to further understand both anatomy and physiology across all animals. F.E. Yates [more memorable author than *Am J Physiol* editor, as he misspelled his own name on this editorial; (249)] argued in his essay "Comparative Physiology: Compared to What?" that the real contribution of Comparative Physiology is inherent in these cross-species comparisons. He made a strong case that body size is the single most important "comparison" that could be used to identify constraints in animal design. Two seminal books subsequently summarized many insights provided by this approach (35, 199).

However, body size is not the only cross-species tool available. Recent studies increasingly exploit engineering and physics approaches to the study of animal locomotion. Unifying principles have emerged from this approach that cross-taxa and modes of movement, again yielding principles that any single species study could not. Examples include analyses of the dimensionless Froude number, a ratio of centripetal force to gravitational force (4), and the Strouhal number, a dimensionless ratio based on the frequency of propulsive movements in swimming or flying animals (222). Likewise, the concept of "Symmorphosis" (220), proposing that tissue and cellular structure should be quantitatively linked to corresponding function is also another unique contribution from a comparative approach.

As the subject of either "muscle physiology" or "animal locomotion" alone is well beyond the scope of any (reasonable) review we attempt here to stay true to these comparative approaches. Hence, this contribution is not a comprehensive analysis of skeletal muscle structure and function or of animal locomotion, but is rather a quest for unifying principles that emerge only from a comparative approach. Rather than a compilation of facts, we have written this review around the goal of identifying common principles in animal locomotion. We envision these principles as the starting points for further inquiry.

Muscle as the Motor of Locomotion

Whether we consider the smallest of animals or the largest, whether an animal swims, crawls, runs, or flies, muscle is the motor that generates the forces required for independent motion. Across taxa, striated muscle retains several common design features that dictate much of its function. This is consistent with the hypothesis that striated muscle evolved very early in metazoan evolution, in an ancestor to all bilateral groups as well as jellyfish (Cnidaria and Ctenophora), all of which possess striated muscle in their sessile forms (200). In response to selective pressure, animals have evolved a wide array of solutions to locomotor challenges by alterations in the

cellular makeup of skeletal muscle that permit specialization for a wide range of functional strategies. Further, muscle has been coopted for nonlocomotor tasks, such as noisemaking (111, 192, 203) and thermoregulation (21) that have largely reduced or eliminated its more traditionally considered role as a force producing tissue.

While the extant functions of muscle are extremely broad, the diverse suite of muscle functions is in all instances accomplished by just three main components: myofilaments, sarcoplasmic reticulum (SR) and mitochondria. As the relative cellular volume density of any one of these components changes so too must one or both of the others, generating a predictable trade off in cellular composition that reflects function. A three-dimensional plot of these three components of muscle cellular structure reflects the tradeoff inherent in any modification as each is expressed as a percent of total cellular volume (Fig. 1). The composition of every skeletal muscle cell must be located on a single plane of this three-dimensional plot. For example, if the percentage of cellular volume is occupied by myofilaments is increased, the remaining space for mitochondria and/or SR must perforce be decreased.

While a specific muscle cell (fiber) was once thought to be genetically programmed for its specific task [and erroneously thought and taught to be so for many years, see e.g., reference (75)], experiments beginning over 50 years ago (32) have documented the remodeling of the cellular architecture of the mammalian muscle fiber that is seen in response to shifts in patterns of muscle use as well muscle stretch, nutritional status and hormonal influence [see references (34, 89, 146)]. This phenotypic plasticity of skeletal muscle is likely an ancestral feature of skeletal muscle as (postdevelopmental) shifts in muscle mass, contractile and/or metabolic properties have been documented not only in mammals, but also in birds (174), reptiles (193), fishes (105), and crustaceans (155, 176).

During locomotion, skeletal muscle accomplishes an astonishing range of functions, with force production for acceleration, deceleration, and stabilization. Additionally, force production may be sustained for long-duration activity or maintenance of body position or very brief during rarely used bursts of high power. These demands vary spatially among muscle groups or temporally during the life of an organism but are met by surprisingly simple structural modifications. Within an organism's life, the relative contribution of each component to the total cellular volume is malleable in response to changing demands, accounting for much of the diversity in muscle function. As demands change, muscle responses include not only quantitative alteration of the volume density of the components of the myocyte, but also qualitative changes as different protein isoform expression lead to modulation of function. Thus, a shift of each of the three major components of the myocyte results in a shift of function via phenotypic plasticity within the organism as well as across evolution. These shifts in structure begin to inform our understanding of how muscle is tuned to the mechanics of locomotion.

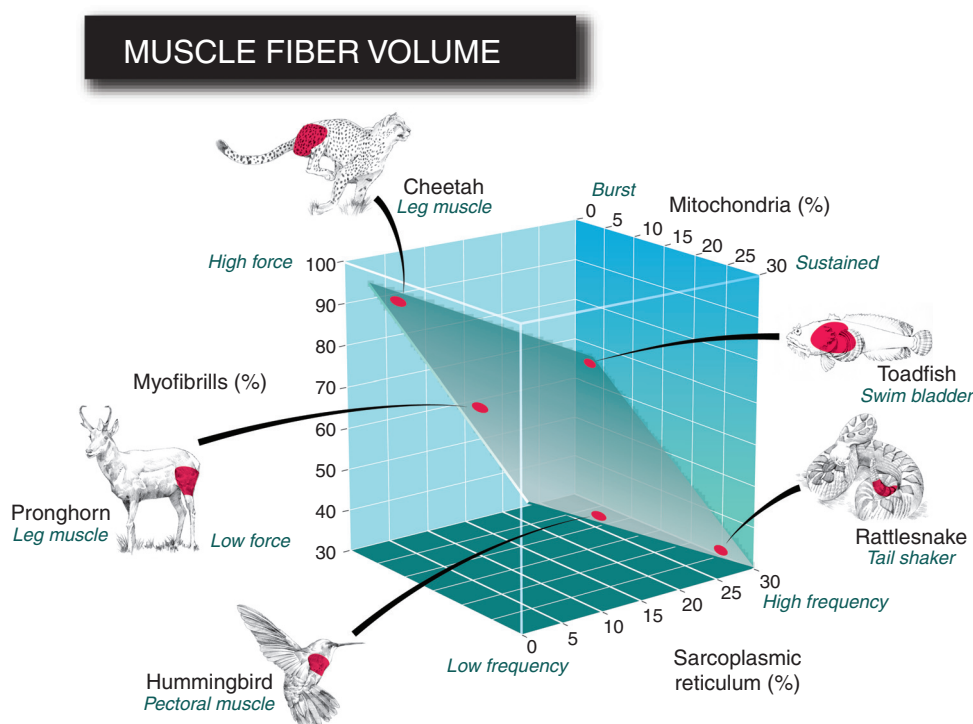


Figure 1 The volume of a muscle fiber is composed of three (space occupying) structural components: myofibrils, mitochondria, and sarcoplasmic reticulum (SR). Because these structures collectively fill the fiber volume, on this three-dimensional graph every muscle fiber must fall somewhere on a single plane. As the contractile proteins compose the myofibrils, any increase in mitochondria or SR (i.e., demand for sustained performance or high-frequency muscle use) must be at the cost of reduced force (and power) production. The wide range of muscle functions, from posture to ballistic use to noise making are all possible not because of unique structures, but rather a rearrangement of these structural elements. We have included a selection of vertebrate muscle examples that occupy different regions of this three-dimensional space; see text for discussion [modified, with permission, from reference (133)].

Structure—function relationships of muscle

Myocyte constituents: myofilaments

The interaction of sarcomeric myofilaments generates the force necessary for movement. The force applied along a given vector is dependent upon the cross-sectional area of the active muscle. During locomotion, there is considerable effect of gearing as limb configurations (and thus force application) change with skeletal lever system movement (37). In addition, the area of active muscle can be increased by increasing the pennation angle (i.e., variation in the origin and insertion of muscle relative to a joint) at which a muscle is oriented to that vector. With increasingly oblique angle of pennation, more muscle can be packed into a given cross-section; however, the force generated by each myocyte is reduced. This trade-off maximizes force capacity at a pennation angle of 45° (2). During locomotion, gearing within the muscle tissue also occurs as a result of changes in fascicle thickness or rotation (226). Significantly, muscle gearing is variable depending upon force generated. At high forces, fibers operate at a low gearing, enhancing force production, while at lower forces, high gearing favors velocity of contraction (9). Musculoskeletal architecture is largely set during develop-

ment and modification of pennation angle is generally not an option available during an organism's lifespan. However, at least in humans undergoing high resistance strength training, pennation angle can vary to some extent with altered use and contribute to force production (62).

The sarcomeres, composed of interlaced contractile filaments within the individual myocytes, are the functional units of muscle contraction. The thick and thin filaments each consist of several proteins, principally myosin heavy chain (MHC) and actin, along with regulatory proteins troponin and tropomyosin, respectively. The composition of the sarcomeric proteins is continually remodeled, with both MHC (described below) and actin proteins regularly replaced (164). Interaction of the actin and myosin filaments during cross-bridge cycling generates the force necessary for motion. How this force is transmitted to the environment varies widely, reflecting both myofilament diversity and variation in muscle-activation patterns. The frequency with which cross-bridge cycling occurs determines the velocity of contraction. Force and velocity determine the majority of muscle mechanical properties. Myocyte force production is modulated by muscle length, that is, the degree of stretch (and thus myofilament overlap) as well as the frequency and duration of stimulation received from

Table 1 Dimensions and Units for Locomotor Variables

Variable	Equal to:	Dimensions	SI Units
Velocity	Distance time ⁻¹	LT ⁻¹	m s ⁻¹
Acceleration	Velocity time ⁻¹	LT ⁻²	m s ⁻²
Force	Mass acceleration	MLT ⁻²	Newton (N)
Work	Force distance	ML ² T ⁻²	Joule (J)
Power	Work time ⁻¹	ML ² T ⁻³	Watt (W)
Stress	Force area ⁻¹	ML ⁻¹ T ⁻²	N m ⁻²
Strain	Length change initial length ⁻¹	Dimensionless	–
Strain Rate	Strain time ⁻¹	T ⁻¹ (frequency)	s ⁻¹
Volume-specific work	Work volume ⁻¹	ML ⁻¹ T ⁻²	J m ⁻³
Volume-specific power	Power volume ⁻¹	ML ⁻¹ T ⁻³	W m ⁻³

The dimensions of each variable are noted using the mass (M), length (L), time (T) system. SI units are noted as well. Note that volume-specific work or power can be expressed as mass specific by including a correction factor for tissue density (1.06 g cm⁻³). The SI units become J kg⁻¹ or W kg⁻¹, respectively, while the dimensions are transformed to L²T⁻² for mass-specific work or L²T⁻³ for mass-specific power.

motor neurons. At optimal length and motor neuron stimulation frequency, each cross-bridge interaction generates about 6 pN of force during an isometric contraction (173). Maximum muscle *stress* (isometric force generated per unit of cross-sectional area) is nearly constant across animal groups at approximately 30 N cm⁻² (170). Hill (86) and later Pennycuik (170), remind us of the insights that emerge just by paying careful attention to the dimensions of the variables of muscle mechanics (Mass, M; Length, L; and Time, T) and we list pertinent variables in Table 1. For example, as force (mass acceleration) has dimensions of MLT⁻², stress therefore has dimension of ML⁻¹T⁻².

For a muscle of a given cross-section to generate greater isometric force production, the available cross-bridges must be increased [e.g. reference (22)]. Within the cell, changes in the proportionate volume density of myofilaments within the sarcoplasm alter the maximal stress. However, the degree of modification available for an organism is rather small as the “default” myofibrillar density is high. For example, human weight training is reported to lead to no change (138) or a small increase (139) in myofilament volume density. However, across evolutionary time scales, the rattlesnake tail-shaker muscle, a noisemaker muscle that generates very low force (150), has undergone a shift from a characteristic muscle composition of about 90% to about 30% myofilament volume density, among the lowest measured [Fig. 1; reference (192)]. Isometric force can also increase without changing the proportion of myofilaments within a cell; hypertrophy of the cell will increase cross-sectional area. Thus, increasing muscle size alone will increase the magnitude of isometric force production in direct proportion to the increase in cross-sectional area; muscle hypertrophy is a response to resistance type of activity that can occur throughout life (100). Maximum isometric muscle stress is largely a fixed property of muscle sarcomere density across taxa; however, isometric contractions do not result in animal movement. Although these contractions are important in stabilization of body elements during locomotion (143), without movement, neither work nor power are produced. Work can be performed indirectly if force gen-

erated isometrically can be transferred to an elastic element (182).

Muscle often changes length when active. To perform work *directly*, muscle must produce force while actively shortening, a concentric contraction. While stress may be augmented by a quantitative increase in the cross-section of myofilaments available for force generation, variation in the rate at which myofilaments move is associated with qualitative differences in the ATPase activity of the MHC (12, 188). Within a given species, MHC proteins are expressed as several distinct isoforms and the rate of ATP cleavage and thus the maximal muscle contraction velocity ($V_{\max}$) varies with each isoform (24, 196). Across species, these rates also depend on body size such that both rates are highest in small animals (Table 2). Muscles also vary in length, with sarcomeres arranged in series. Each sarcomere undergoes similar length changes during activity. Therefore, absolute velocity is the sum of individual sarcomeric velocities. *Strain* (the dimensionless ratio of shortening distance divided by resting muscle length) serves as a measure of relative shortening and *strain rate* (calculated by dividing the velocity of shortening by the resting length) represents the strain per unit time (Table 1). As the ratio of velocity divided by length, strain rate is a pure rate, with dimensions of T⁻¹.

In addition, muscle is commonly activated during stretch, leading to lengthening contractions (often called eccentric contractions), during which contraction velocity is negative and thus work is done on the muscle. Importantly, as force and distance are vectors (rather than scalars), F and D must be quantified by both their magnitude and direction. As a consequence, when a muscle produces force while shortening, $F \times D$, the result is work, and while lengthening, the negative distance resulting in “negative work”, $F \times -D = -W$ (1).

Eccentric contractions permit muscles and tendons to act as springs, storing, and recovering elastic strain energy. In addition, muscle can function as a “shock absorber” during lengthening contractions, for example, moving downhill, stabilizing the body during the deceleration phases of each cycle. The work done on the muscle during a lengthening

Table 2 Stride Frequencies in Sustained Locomotion: Running, Swimming and Flying

Variable	Units	<i>a</i>	Exponent (<i>b</i>)	<i>r</i> ²	Source
Running:					
Quadrupeds at trot-gallop transition	s ^{−1}	4.48	−0.147	0.99	(81)
Quadrupeds at trot-gallop transition	s ^{−1}	4.19	−0.150	0.87	(80)
Quadrupeds at maximum sustained gallop	s ^{−1}	4.7	−0.162	0.88	(80)
Quadrupeds, preferred trot frequency	s ^{−1}	3.35	−0.130	0.84	(80)
Bipeds (birds, humans) at top speed	s ^{−1}	4.9	−0.178	0.98	(66)
Rate of muscle shortening (mammals)	s ^{−1}	2.93	−0.23	–	(131)
Swimming:					
Fish: pectoral fin beat frequency	s ^{−1}	3.07	−0.135	0.90	(55)
Fish: tail beat frequency	s ^{−1}	–	−0.17	–	(6, 20)
Flying:					
Insects	s ^{−1}	8.95	−0.21	0.23	(50)
Birds	s ^{−1}	3.87	−0.175	0.48	(31)

Allometric power law equations are presented for a number of locomotor frequencies. The units are pure frequencies in all cases. The equations are in the form $Y = aM^b$; where “ Y ” is the variable of interest, “ a ” is the intercept, “ M ” is the body mass (kg) of the animal, and “ b ” is the exponent, derived from the slope of the resultant line when data are plotted on a log-log graph. References for each equation are noted.

contraction may be subsequently captured, stored, and used to produce work; allowing the muscle to uncouple force production and work. Eccentric contractions are an integral part of locomotion, as recovered stored elastic strain energy reduces the work done by the muscle (132). Novel, high-force eccentric contractions have been demonstrated to cause significant muscle damage, avoidable when the organism gradually increases its use of this activation mode (61). This function may be contributed by the protein titin. Although it was one of the last proteins discovered (144, 228), titin is the largest known protein and the third most abundant protein in skeletal muscle, spanning the entire half-sarcomere from M-line to Z-disk [see reference (74)]. The structure of this enormous protein is ideally suited as an elastic structure within the muscle and its passive elasticity has been the subject of considerable interest especially in the heart [for review, see reference (127)]. Titin has been suggested to function as a spring in active skeletal muscle as well (83, 132, 179), an idea that has gained considerable traction recently with the discovery of calcium-dependent increase in titin spring stiffness (84, 122).

Thus, the capacity for myofilament force production is a function of both the magnitude and rate of force generation. These parameters are determined by the quantitative and qualitative variation in the properties of the myofilaments. The extant variation in myofilament properties reflect the wide range of demands that diverse animals must face. Evolutionary forces have selected force production in part by varying the volume density of myofilaments as demonstrated by the cheetah in Figure 1. In contrast, muscles used for noise making require little force, thus the myofilament volume density of the rattlesnake tail shaker is among the lowest yet recorded (Fig. 1). Concurrent with quantitative shifts, qualitative variation in shortening velocity is responsible for large differences in the functional mechanics of muscle and contribute considerably to variation in power output as described in Section “Muscles in Movement,” below.

Myocyte constituents: Sarcoplasmic reticulum

Neuronal excitation of muscle is coupled to force generation (excitation contraction coupling) via calcium release from the SR. The rates of muscle activation and relaxation are both directly related to the kinetics of calcium release and reuptake. Indeed, with the space-saving exception of asynchronous muscle, frequency of activation in cyclically used muscle is closely correlated with the volume density of SR within the muscle fiber (Fig. 2). The absolute cycle frequency is determined by the properties of the SR and must be coordinated with the maximal contraction velocity; high frequencies require fast myosin cross-bridge cycling. Across taxa, cellular

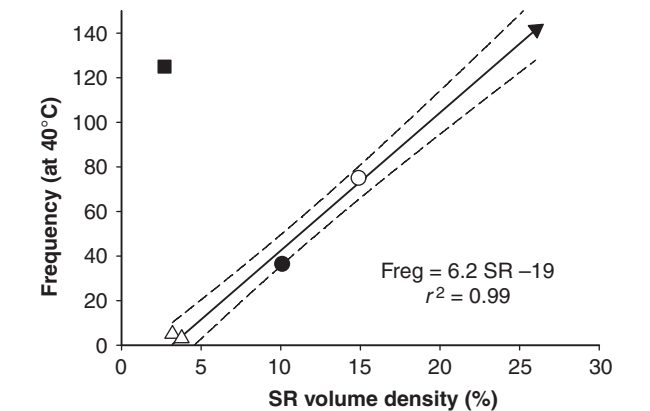


Figure 2 For muscles to function at different operating frequencies calcium cycling kinetics must be tuned to frequency of muscle use. As a consequence, there is a consistent and predictable relationship between frequency of muscle use and the volume of the muscle fiber devoted to sarcoplasmic reticulum (SR). In this figure, when necessary we have normalized data to a muscle temperature of 40°C using the Q_{10} reported for these specific animals. Data are from human limb muscle (95), guinea pig limb muscle [(58, 64); both mammals open triangles], hummingbird flight muscle [filled circle, (240)], cicada noise-maker muscle [open circle, (110)], and rattlesnake noisemaker muscle [filled triangle, (192)]. In contrast, asynchronous insect muscle achieves high frequencies with a low volume density of SR [square, (110)].

volume densities of SR correspond to expectations from function. When the rates of contraction are very high, such as in hummingbird flight muscle or rattlesnake tail-shaker muscle, SR volume densities are also quite high (Fig. 1). Conversely, for animals such as the pronghorn antelope, in which muscle cycle time is not particularly high, SR volume density is correspondingly low (Fig. 1). However, data from human exercise training show no change in this parameter (7,95) and we are not aware of data examining the effect of altered activity on SR volume density in other animals.

Calcium cycling influences contraction frequency in a cyclically used muscle via both activation and relaxation rates, each of which is modulated by quantitative and qualitative variation in SR structure and function. Muscle activation follows calcium release from the ryanodine receptors (RyRs), an integral membrane protein of the SR. Although muscle fibers within several vertebrate species, with large differences in contraction velocity, express fiber-type specific isoforms of the RyR, there is no evidence that these isoforms influence the rate or quantity of calcium released from the SR (211). Thus, their functional relevance remains unclear. Calcium release is a passive process, dependent upon excitation of the sarcolemma to initiate release of SR calcium stores. The high-concentration gradient from the SR to the cytoplasm ensures that calcium flows quickly out of the SR, into the sarcoplasm and to the sarcomeres. With greater volume density of SR, the diffusion distance is reduced, thus increasing both the “on” and “off” rates (188). Following diffusion to the myofilaments, calcium binds to troponin C (TnC) and a conformational change permits cross-bridge cycling. TnC is expressed in both fast and slow isoforms whose expression corresponds to the MHC expression (161). While the affinity of TnC for calcium varies with fiber-type specific isoform, this has no apparent effect on muscle activation time as the kinetics of cross-bridge formation are slower than the calcium transient (185).

Upon cessation of nervous stimulation, calcium is sequestered into the SR and muscle relaxes. Relaxation time is considered to be the primary determinant of contraction frequency (185). The calcium diffusion distance is a function of the density of SR. In addition, the time required for relaxation is dependent on the sarco-endoplasmic reticulum calcium (SERCA) ATPases, responsible for pumping calcium from the sarcoplasm into the SR (26). There are three *SERCA* genes in mammals and multiple isoforms of these genes are expressed following alternative splicing events (26). Up to 90% of the proteins associated with the SR itself may be SERCA pumps, expressed in fiber-type specific patterns. The major isoforms are associated with either fast muscle (*SERCA1*) or slow muscle (*SERCA2*). *SERCA* function is modulated by accessory proteins including phospholamban (PLB), which decreases *SERCA* affinity for calcium (171, 214). Both *SERCA2* and PLB are induced by chronic muscle activity, concordant with a fast-to-slow fiber-type shift (97), qualitatively altering calcium reuptake in place of quantitative change in the volume densities of SR. *SERCA* splice variants modulate the PLB-

binding sites and thus potentially modulate muscle frequency (26), although this has not been investigated in the context of adaptation to altered use patterns. Relaxation rates, determined by the rate of removal of calcium from the sarcoplasm, are thus influenced by the properties of the SR, in particular its volume density, as this determines diffusion distance to myofilaments, and by the quantity as well as the qualitative nature of the SERCA pumps. In addition, using mutant versions of TnC with altered calcium affinity, several studies have demonstrated that the rate of calcium release from TnC can alter relaxation rate (137). Like SERCA, TnC is expressed in fast and slow muscle specific isoforms and its expression is responsive to altered activity level (152). Whether calcium reuptake or calcium detachment determines relaxation kinetics depends upon fiber type, such that the fastest fibers are more dependent upon the affinity of TnC for calcium while relaxation time of slower fibers is primarily dependent upon SERCA activity (185). In the superfast noisemaker muscles of the toadfish, very large amounts of the calcium buffering protein parvalbumin are found, permitting the SERCA pumps to operate at lower than expected rates yet maintain a fast relaxation rate (96, 186). These superfast noisemaker muscles exemplify a unique solution to the problem of high frequency. They require significant rest periods between bouts of activity for calcium clearance, thus avoiding the need for high-sustained power input. The apparent fatigue is due to a calcium handling limitation as parvalbumin becomes saturated, rather than an energetic limitation (185). As the frequency of cyclic use increases, together with increased volume density of SR, increased expression and isoform shifts in SERCA pumps and TnC coordinate to reduce the duration of the calcium activation of the cross-bridges and speed contraction and relaxation.

Myocyte constituents: mitochondria

Aerobic energy production by the mitochondria supplies the majority of the ATP-consuming processes. Sustainable locomotion is defined by the ability of the mitochondria to provide sufficient energy equivalents for cross-bridge cycling as well as ion transport. When muscle activity varies, whether in tissues within an organism, or among the same muscles of different organisms, mitochondrial function must be sufficient to supply that demand or activity duration will necessarily be limited. Estimates of the energetic costs during muscle contraction are about 70% for cross-bridge cycling and 30% for Ca^{++} reuptake (13, 177). These values assume that the contribution of the Na^+/K^+ pump is low during contraction, based on estimates of ATP cost and the observation that sodium accumulation requires that the pumps remain active for extended periods (13). This proportion in costs is remarkably similar in a range of vertebrates studied, including fish, amphibians, and mammals, with a large range in the volume densities devoted to myofibrils and SR (13). This proportion in energy costs is even maintained during contractions in the superfast swim bladder muscle of the toadfish in which the volume density of

SR is very high (188), although this must be qualified by the observation that much of the SR pumping costs are deferred to periods of inactivity (185).

In contrast to contractile and calcium-handling proteins, mitochondrial proteins are not expressed in fiber-type or temperature-dependent specific isoforms. Instead, mitochondrial ATP production is directly related to the temperature and volume density of mitochondria within the muscle tissue, regardless of fiber type, likely a legacy of mitochondrial evolution. Examples of quantitative shifts in mitochondrial volume density abound as expansion and contraction of this pool may represent the most plastic of muscle cellular components. Among noisemakers, rattlesnake tail-shaker muscle may be continuously active for several hours and has extremely high mitochondrial volume density, in stark contrast to toadfish swim bladder muscles that are only intermittently active. Pronghorns and hummingbirds are capable of sustaining high levels of metabolic power output and also have correspondingly high mitochondrial volume densities in their locomotor muscles. In contrast, animals specialize for short, burst activity, have no need to a continuous supply of ATP, thus the mitochondrial volume density in cheetah muscle is greatly reduced [(244); Fig. 1]. Human training leads to similar outcomes, with expansion of the mitochondrial volume following sustained, endurance activity (92, 95), and contraction with intermittent, resistance exercise (7, 138, 139). Across species, mitochondria appear to possess very similar properties even though rates of energy use in different individual muscles may vary many-fold. Demonstrating constancy of mitochondrial function *in vivo*, mammals ranging in size from mice to horses have nearly identical rates of maximal mitochondrial respiration [$\sim 4\text{--}5 \text{ mL O}_2 \text{ cm}^{-3} \text{ min}^{-1}$; (92, 237)], independent of body size. The higher mass-specific metabolism of smaller animals, whether at rest (116) or at maximum aerobic capacity (218) is primarily attributed to mitochondrial volume rather than any alteration in the nature of the mitochondria (145), although inner membrane proton leak varies with body size (99).

In ectotherms, variation in temperature can lead to profound differences in enzyme functional rates, including mitochondrial respiration, as eloquently described by Hochachka and Somero (87). It appears that the depression of mitochondrial respiration is a direct effect of temperature. When temperature effects are controlled, mitochondrial respiration rates are remarkably constant in a wide range of animals including reptiles, amphibians, and mammals, with highly divergent lifestyle and activity level (192). In those organisms with the coldest body temperatures, for activity to be maintained, the volume density of mitochondria in muscle cells may exceed 50% to ensure sufficient supply given a Q_{10} of approximately 2.5 (106). Thus, in cold temperatures the maintenance of ATP synthetic capacity is dependent upon dramatic expansion of the cellular volume devoted to mitochondria.

One striking exception to the observed constancy in mitochondrial respiration rate per unit volume is found in small flyers, including hummingbirds and insects (192). In these

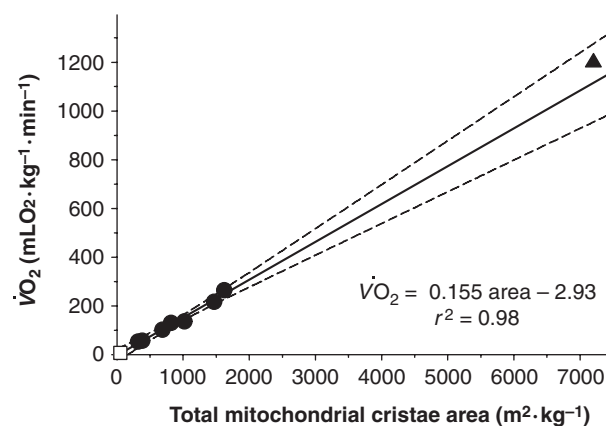


Figure 3 Maximum ATP synthetic capacity (and thus oxygen uptake) in skeletal muscle is a direct function of mitochondrial capacity to process reducing equivalents. In mammals, this seems to be a simple function of muscle mitochondrial volume. With variation in cristae density across all taxa, oxygen uptake most closely correlates with the total quantity of inner mitochondrial membrane surface area. The maximum rate of oxygen uptake per unit of inner mitochondrial membrane area is nearly constant when all data are normalized to a common temperature of 40°C (using a Q_{10} of 2.2) in rattlesnakes (square) mammals (circles) and hummingbirds (triangle).

animals, the mechanical power demands of flight are very high, requiring high myofilament volume density. Yet, ATP demand during flight is also among the highest yet measured (159, 209, 210, 240), thus requiring high ATP synthetic capacity as well. Mitochondrial volume density appears to be insufficient to supply ATP to maximally working muscle if assumed to operate at the “standard” respiration rate. This suggests that the trade-off between capacity for mechanical and metabolic power has reached a design constraint, such that further expansion of the mitochondrial pool would excessively diminish the capacity for myofilament force production. This spatial limit has been circumvented as mitochondria from these animals consume oxygen at about double the rate described above. This is accomplished by a doubling of the density of the inner cristae membranes, which house the electron transport chain and ATP synthase (17, 209, 210, 240). When the rates of oxygen consumption of mitochondria are compared to total mitochondrial cristae surface area, there is a remarkably constant relationship among many organisms (Fig. 3). When these rates of oxygen consumption are calculated per unit of cristae surface area, and normalized to a similar operating temperature, hummingbird mitochondrial oxygen uptake is identical to other animals at about 70,000 molecules of oxygen per minute per square micron of inner mitochondrial membrane at 40°C (135).

Overall, mitochondrial respiration rate appears to be remarkably uniform and the importance of constancy in mitochondrial function is supported by evidence that in humans, inborn errors in mitochondrial function can directly cause severe pathology (16). However, Jackman and Willis (101) found evidence of fiber-type specific functional variation in substrate preference in oxidative and glycolytic muscle of the rabbit. Generally, although aerobic capacity may vary

many-fold, at an activity level requiring any given percent of that capacity, mitochondrial respiration results in consumption of fatty acid and glucose fuels in a remarkably uniform proportion (29,231). Studies in humans and rodents have also revealed that a high-fat diet can shift this relationship toward greater reliance upon lipid fuels (104,156). These substrates that fuel oxidative ATP production are often in direct contact with the mitochondria. Both intracellular glycogen and lipid stores can occupy variable amounts of cellular space and often accurately reflect the functional nature of the muscle cell, such that lipid stores are associated with highly oxidative muscle while glycogen tends to be more abundant in glycolytic muscle. We discuss these fuels in conjunction with tissue level supply in Section “Energetics of Locomotion.”

Taken together, the above suggests that the function of mitochondria, perhaps owing to a common ancient origin (191), is remarkably conserved. Unlike the myofilaments or the SR, the mitochondrial compartment of skeletal muscle has evolved little qualitative variation in the muscles of animals with different locomotor styles or within different fiber types. With notable exceptions, altered demand due to changing locomotor activity is met with variation in the volume density of mitochondria of the affected fibers.

Skeletal muscle fiber types

The most obvious distinguishing feature of skeletal muscle is its color. Whether a muscle appears “Red” or “White” reflects the concentration of pigmented proteins within the tissue (especially myoglobin). Beyond color, histological and immunohistochemical techniques permit differentiation of myocytes into characteristic fiber types based on myofilament and metabolic properties. These types are best defined in mammals (27), but are commonly identified in other taxa as well (11,39,197,223). While muscle fiber types are somewhat theoretical, as distinguishing properties exist along a continuum, they do represent broad patterns that are useful in discussion of muscle properties related to locomotion. In vertebrate locomotor muscle, fiber types are broadly demarcated as being either slow or fast, with fast fibers being further differentiated into two to three subdivisions. Differentiation of fiber types is based upon MHC isoform. Although only one MHC protein isoform is typically expressed in a single myocyte, hybrid (or transition) fibers may express more than one MHC protein as fiber-type shifts (162) or, multiple MHC isoforms may be stably expressed along the length of an individual fiber (251). Fiber types are further differentiated based upon metabolic properties (oxidative or glycolytic) that comprise a continuum of phenotypes ranging from primary reliance on glycolysis to primary reliance on mitochondrial respiration. In general, oxidative properties are highest to lowest, and contractile velocity slowest to fastest along the continuum of type I (slow-oxidative), type IIa (fast-oxidative), type IIc/x (fast-intermediate), and type IIb (fast glycolytic). Although there is little difference in peak stress, variation in velocity has profound effects on work rate or power such that

fast muscles are able to generate higher power. The trade-off being that during comparable activation, slower muscles operate more efficiently (246). Along this continuum, metabolic properties are not distinctive, instead existing in a nearly continuous range within a given organism (172). Across taxa, muscle metabolic properties vary widely from highly glycolytic cheetah leg muscles to highly aerobic hummingbird flight muscle.

Variation in the rates of force development is associated with qualitative differences in the myosin ATPase activity (12) which determines maximal contraction velocity of the tissue (188), and hence a suite of mechanical properties of the muscle (see below). Thus, fiber type serves primarily as a predictor of these functional properties. However, it is worth noting that muscle fibers classified as identical “types” may have widely differing properties in different animals. For example, the slowest contracting fibers in a small mammal such as a shrew may still be about five times faster than the fastest fibers in a larger mammal, such as an elk (213). See reference (204) for further, excellent treatment of skeletal muscle fiber types.

Locomotion and molecular regulation of muscle structure

Development of locomotor properties of muscle

The specific myocyte composition of muscle usually reflects that muscle’s locomotor demands. This process builds on developmental patterns as innervation is established and continues throughout life. The pathways responsible for establishing skeletal muscle phenotype as well as the mechanisms responsible for responses to the demands of locomotion are becoming delineated. Developmental processes appear to reflect the character of adult muscle as early innervation begins the process of matching the cellular composition and functional phenotype of developing muscle to the mechanical and metabolic demands exerted later in life.

The matching of muscle structural properties to the nature and intensity of muscle tasks begins during early development. In both chick and mouse embryos, populations of myoblasts form, each one of which is destined to form skeletal muscle possessing either a fast or slow fiber type (63), although there is evidence that this program can be overridden by local factors (112). These primary fibers form before contact is established with motor neurons and they express embryonic forms of MHC proteins that can be assigned to either fast or slow myosin gene families (208). During fetal stages of development, these fibers are innervated and converted to secondary fibers—a process that is characterized by a regular transition of MHC isoforms (11,242)—although prevention of motor innervation prevents this transition. Initially, as muscle fibers develop, the embryonic myosin is replaced first by fetal isoforms and finally adult isoforms, beginning with fast and progressing to slow MHC. While the majority of the fiber-type diversity is established early in development (208), after development, changes in activity patterns as well as hormonal influences lead to further shifts

throughout life. Unlike embryonic muscle, expression of slow myosin isoforms in adult fibers is dependent upon innervation, and thus is likely a function of the demand placed on the muscle. It appears that the isoforms associated with fast muscle are the default condition and activity is responsible for modification of muscle structure (72).

As mitochondrial volume density is lowest in muscles that are used infrequently, following development, as frequency of activation increases, mitochondrial biogenesis expands the volume density through an active fissioning process (163). Thus across taxa, mitochondrial structure-function relationships are maintained such that supply and demand are closely matched. Both during development and in adults, the regulation of this mitochondrial biogenesis and all organellar respiratory function is a complex system involving coordinated action of both the nuclear and mitochondrial genomes; both genomes generating mRNA for protein subunits of nearly every complex within the respiratory chain [reviewed in reference (113)].

Regulation of adult plasticity of muscle in response to locomotor demands

Adult plasticity of myocyte structural and functional properties occurs in response to numerous convergent signaling pathways, which are often activated by changes in muscle activity (14,91). These include: sustained elevations of intracellular calcium, direct detection of mechanical stresses via mechanoreceptor pathways, and energy sensing via elevation of the AMP concentration. These pathways explain the link between locomotor demand and shifts in muscle properties that facilitate the matching of tissue properties with demand.

In addition to activating cross-bridge cycling and modulating spring stiffness through titin binding, calcium acts as a second messenger in many signaling pathways. In muscle, calcium release activates a number of pathways that lead to altered gene expression and cellular composition in a process known as excitation-transcription coupling. One of the best-understood factors responding to this calcium signal is the protein phosphatase calcineurin. Calcineurin activity is initiated by sustained elevations of intracellular calcium typical of slow muscle activity patterns, leading to activation of a family of transcription factors (178). The promoters of slow muscle-specific isoforms of several contractile proteins are activated by these proteins implicating calcineurin signaling in an activity driven shift toward slow fiber types (40). Mice overexpressing constitutively active calcineurin in skeletal muscle have more slow fibers (52, 157, 248) while genetic ablation of calcineurin reduced the proportion of slow fibers (166, 167). Further, calcineurin activates expression of genes responsible for fatty acid oxidation and mitochondrial biogenesis (195, 247). Elevated intracellular calcium concentration also participates in coordinated regulation of MHC isoform expression in an elegant system regulated by calcium/calmodulin-dependent protein kinases that modify the histone environment of these genes. In mammals, *MHC* genes are located sequentially on a

single chromosome such that promoter activation leads to expression of one isoform along the sense strand and inhibition of expression of another isoform along the antisense strand (165, 181).

Locomotor activity consumes energy equivalents and sensing of perturbation in the cellular energy status is accomplished by the adenylate kinase system (205). The energy status of the cell is tightly regulated and accumulation of AMP, or reduction of the ATP/(ADP+AMP) ratio, activates AMPK, a protein kinase system involved in regulating this status (250). AMPK action stimulates expression of numerous pathways responsible for ATP supply, including glycolysis and fatty acid oxidation, while inhibiting anabolic pathways (205), leading to long-term changes in the ability to supply ATP to working muscle (36).

Muscle activity also stimulates muscle growth through increased expression of a variant of insulin-like growth factor I (IGF-I) derived from alternative splicing (70). This variant, termed mechano growth factor, is synthesized in response to mechanical stimuli as well as muscle damage and acts in a paracrine fashion, leading to increased mass of both active muscle and surrounding tissue (71). In addition to quantitative changes in mass, continuous mechanical stretch is capable of stimulating a fast-to-slow MHC isoform shift in the absence of any neuronal excitation (243).

As increased frequency of locomotor activity leads to elevation of both intracellular calcium concentration and AMP, these signaling pathways are consistent with induction by sustained muscle activity, stimulating development of a muscle phenotype with high oxidative capacity and thus high endurance. In those muscles for which high power is favored, high muscle stresses generated during locomotion stimulate regulation of MHC isoforms, hypertrophy of the tissue and reduction of mitochondrial volume density. In either case, it is the activity of the tissue, acting together with the genome, that tunes structure to function.

Emergent principles

Most of the volume (>95%) of a muscle cell is composed of just three structures with the many nuclei and stored fuels accounting for the remaining volume. For any one component to be quantitatively altered there must be a proportionate change in the relative amount of one or both of the others. Beginning in development and throughout adulthood, muscle structure is dynamically regulated to ensure that it is tuned to accomplish specific tasks by shifting the proportion of cellular space devoted to each of these primary components, fulfilling the predictions of adaptive linking of structure and function (235).

Understanding of skeletal muscle function has thus moved from a paradigm in which muscle properties were assumed to be fixed during development to one in which the importance of phenotypic plasticity is a central theme; “muscle is what muscle does.” However, it is also appreciated that plasticity in response to changing activity is not absolute. It is uncommon for increased use to transform type II fibers

into type I (204) although the nearly complete transition has been accomplished experimentally (93). Similarly, complete inactivity (as in the spinal isolation model) leads to atrophy with very little change in fiber-type diversity (190). In species that undergo prolonged inactivity (e.g., hibernation), alteration of skeletal muscle properties is minimal (98). Nonetheless, skeletal muscle phenotypic plasticity remains profoundly important in modifying the characteristics of muscle in response to the challenges of locomotion.

Just as evolution cannot be understood without consideration of the interaction between the organism and its environment (128), the properties of skeletal muscle tissue reflect both the genetic program of the species and the specific anatomical location of the tissue, but also the ways in which the organism (and thus tissue) interacts with the environment. As the organism frequently must respond to changing environmental demands, its tissues experience altered demand. The ability of the muscle to deliver (supply) the required functionality relies upon the ability to integrate the demand imposed by behavior, alter gene expression, and undergo tissue remodeling. In contrast to evolutionary processes, this occurs within hours, days, or weeks and throughout the lifespan of the individual. We now explore these properties within the context of locomotion in animals that display the full range of activity behaviors.

Muscle Mechanics During Movement

While different animals move at very different “characteristic” speeds, and even individual animals may have a nearly continuous range of locomotor speeds, most animals have two distinct locomotor strategies/practices. Animals have a cruising speed used for sustained movement. This sustained locomotion, whether of a bird migrating across continents or oceans or a pronghorn moving across the Wyoming plains, seems to have been driven strongly by selection for fuel economy. But most animals also have a very different “ballistic” strategy to move as quickly as possible, for a short duration, often for either escape or capture. As these burst activities are often dire, fuel economy is irrelevant, just as it is in a one-fourth mile dragster, and these movements seem to be driven exclusively to maximize explosive power. However, that is not to say that anaerobic fatigue may be of critical importance even in these burst activities. Thus, it may be the buildup of metabolic waste products (e.g., lactate) that becomes limiting in setting “anaerobic stamina.”

Unifying principles

There has been an apparent paradigm shift in the study of animal locomotion in recent years as the field has moved from explaining the differences among the diverse modes of movement to attempting to find common principles linking them. Thus, unifying principles (23) are now emerging that transcend the obvious distinguishing characteristics of animals in their disparate ways of moving about. Animals vary

extraordinarily in their use of wings, limbs, or full body undulations to provide thrust alone, lift, and thrust in a fluid or use of a scaffold against the ground to combat gravitational acceleration. Importantly, an inclusive view is emerging that, in all instances, whether running, swimming or flying (with endoskeleton or exoskeleton), locomotor muscles of all animals are “tuned” in a manner that links muscle properties to a characteristic frequency of use (48, 153). Various mechanical, energetic, and physical models have been employed to identify and understand the nature of this tuning. For example, general unifying principles of animal locomotion have been identified probing muscle locomotor designs that maintain constant: (i) strouhal numbers [a dimensionless number based on muscle cycle frequency divided by the multiple of amplitude and shortening velocity, see reference (222)], (ii) intrinsic muscle pressure (117), and (iii) the biomechanics of energy storage and exchange with each stride (54), to name just a few recent contributions to our understanding. Emerging is a common general theory of “optimized intermittent movement” (15).

Muscles in movement

Our understanding of muscle force production during shortening and isometric contractions is largely based on the work of A. V. Hill. Hill’s (85) experiments were designed to determine the properties of muscle during single, active contractions. These seminal studies demonstrated the trade-off between the force generated by a muscle and the velocity at which it shortens. The conditions Hill imposed in generating the force-velocity relationship are an often overlooked aspect of these experiments. Hill considered the series elastic elements of muscle to be irrelevant to active contraction and so performed experiments using an afterloaded isotonic contraction model in which maximally activated, isolated muscle contracted against a constant external load. Thus, contraction was isometric until sufficient force was developed, but isotonic thereafter [(85) and see Fig. 4B)]. This mode of activation, convenient in the laboratory, may be rare in natural locomotion. The resultant force-velocity curve (Fig. 4A) has become a staple of physiology textbooks. While this now-familiar relationship still provides valuable insights, it has both limitations and omissions. Among the Hill insights is that muscle work and power are simple calculations. When force and shortening velocities are constant, the length over which muscle shortens, determines the work done, as the product of force and distance is work, and power is the product of force and velocity (see Table 1 for these and other muscle mechanical variables). Quantification of power from the force-velocity curve reveals that maximal power occurs at about one-third of maximal velocity ($V_{\max}$; Fig. 4C).

Of course *in vivo*, both the extent of shortening as well as shortening velocity may vary greatly. For comparison across the wide range of extant body sizes and muscle adaptations for locomotion, size independent measures permit us to determine both how mass-specific power varies among

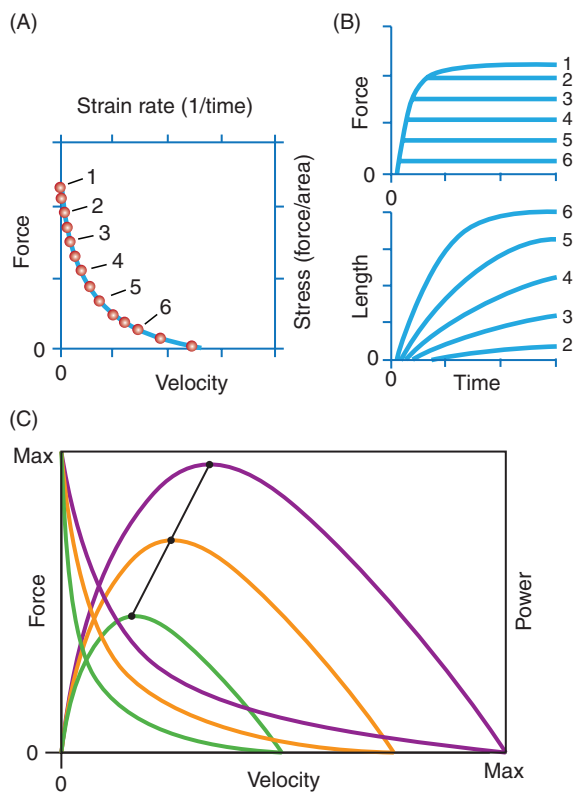


Figure 4 Much of our knowledge of muscle mechanics is traceable to the pioneering experiments by A. V. Hill which form the basis of our understanding of how force, velocity, and power are related in skeletal muscle. In the Hill model, experiments were performed using an afterloaded isotonic contraction model in which maximally activated, isolated muscle contracted against a constant external load. Thus, contraction was isometric until sufficient force was developed, but isotonic thereafter. Using this approach, it can be observed that as the load is reduced, muscles contract at increasing velocity to generate a characteristic force-velocity curve (panel A). The load and velocity of each point that generates this curve demonstrate how as load is reduced, the period during which an isometric contraction occurs is extended before movement begins and the rate of length change increases with reduced load (panel B). The characteristics of this relationship are unchanged when force and velocity are replaced with stress and strain rate (size-independent measures of each variable; panel A). The product of force and velocity are power and the relationship between power and velocity reveals that muscle generates maximal power at about one third of maximal velocity (panel C). While muscles of identical cross-sectional area generate nearly identical force, the rate at which those muscles develop force may greatly affect power generation. The isocline demonstrates the increase in maximal power resulting from increasing maximal contraction velocity (panel C). Panels A and B modified, with permission, from reference (34).

animals, as well as the extent to which contraction velocities affect absolute power generation. Using stress and strain rate in place of force and velocity, one can now generate a curve similar to Hill's that is independent of size (Fig. 4A). As with force and distance or velocity, multiplying stress by strain rate, permits calculation of the volume-specific work or power [(170); Table 1]. Volume-specific power [or mass-specific power if divided by 1.06 to correct for muscle density; (148)], provides a metric for relative power output of any working muscle. When we calculate relative power in this way, the variability in strain rate determines the difference in

specific muscle power output during locomotion. As *absolute* velocity increases, absolute power increases dramatically for any constant strain rate (Fig. 4C).

During sustained locomotion, at all speeds, animals travel at what seems to be a nearly constant velocity, whether on land or moving through a fluid, flying, or swimming. However the muscle machine powering this movement is, in all instances, cyclic. Each stride, wing or tail beat, propels the animal forward, generating thrust by exerting a series of “backward” forces on the environment. Thus, sustained locomotion is characterized by the oscillating contraction and relaxation of locomotor muscles in consistent, cyclic, and reproducible patterns of activation. An outstanding review of this topic by Dickinson et al. (54) examines the mechanics of this cyclic movement from insects to vertebrates to robots. Significantly, this cyclic nature of muscle use in locomotion has provided evolution with an additional dimension not available when muscle contracts in isolation; it allows muscle, with cooperation from its tendon (19), to function as an elastic spring as well as a motor. This added dimension represents an element not represented, or often dismissed (86), in a classic view of muscle mechanics. Consequently, in sustained locomotion our understanding of muscle must expand from the occurrence of an event to the recurrence of a cycle.

Because there is no useful way to fully depict this cycle on the classic Hill force-velocity curve (Fig. 4A), Robert Josephson recognized the need for a new heuristic tool to decipher cyclically contracting muscle. Thus, the work-loop concept is an insightful tool to illustrate muscle mechanics during a locomotor cycle (111). As the starting force and length are identical for repeated cycles, each cycle can be represented on a graph depicting muscle force and muscle length. Further, as the product of force and distance is work, the area contained within the curve of a plot of muscle force versus muscle length is muscle work.

By using as a starting point, the traditional Hill force-velocity relationship, we can construct a virtual work loop, which under these conditions will appear as a simple rectangle (Fig. 5A, left). Starting with the muscle activated in an extended position, force must increase isometrically for a time interval reflecting the combined duration of excitation-contraction coupling as well as motor unit recruitment (I on the plot). As force is increased, eventually the force generated by the muscle equals that of the opposing force. For example, if we examine the knee extensors of a human climbing a stair, the opposing force is body weight. Once this force is achieved, the muscle shortens while producing a nearly constant force (2 on the plot). When the body weight shifts to the contralateral foot, the muscle is deactivated and force drops, initially with no change in length, a period of isometric force reduction (3). Finally, the muscle is brought back to its extended length passively (4) by the actions of muscle antagonists, in this case, the knee flexors. As power is the product of force and velocity, it is also possible to view a single muscle contraction as described above as a “power trajectory,” a useful but underutilized concept introduced by James et al.

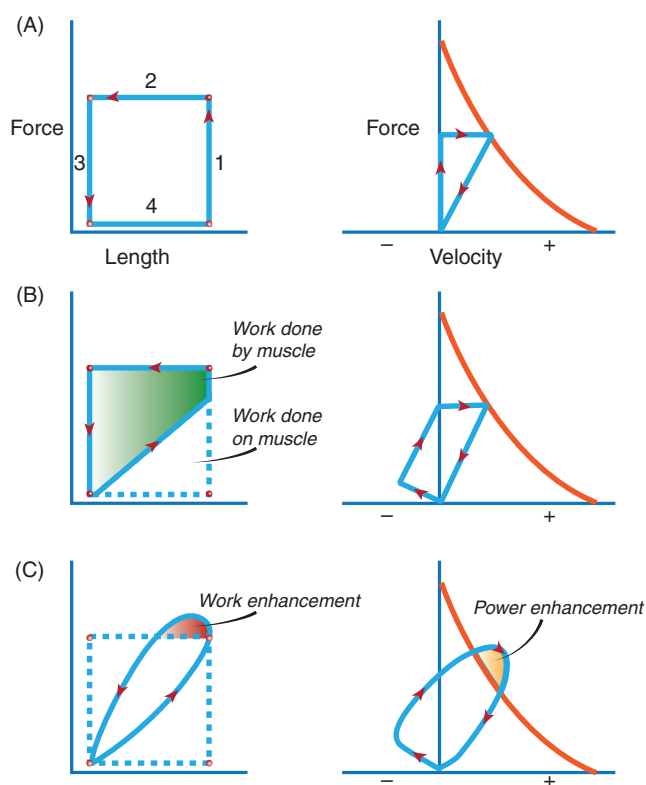


Figure 5 Locomotor muscles are used cyclically. During steady-state locomotion, each stride, wing-beat, or tail-beat re-enacts a stereotyped series of muscle movements with nearly identical force production. Because the product of force and distance is work, the simultaneous tracing of muscle force and length changes during one muscle cycle results in the production of a work loop (110). The work loop reveals insights into *in vivo* muscle mechanics beyond those of traditional Hill mechanics. Likewise, we can also envision the trajectory of power generation *in vivo*, which also departs from a steady-state Hill model. As power is the product of force and velocity, we can determine the power generated at any point during muscle activation. By following a single muscle cycle, it is possible to follow the trajectory of power production. (A, left) If a hypothetical work loop is created for an isolated contraction (as in Figure 4), it appears as a rectangle. Force must increase isometrically (i) until sufficient force is generated for the muscle to shorten isotonically (ii). When the muscle relaxes, force will fall, also isometrically (iii), until the muscle is returned (e.g., by an antagonist) to the starting position. The area of this rectangle is the work done by the muscle. (A, right) For the same contraction, the power generated by the muscle is depicted. A power trajectory for the same isolated contraction is overlaid on a Hill plot of force and velocity. Again, force must increase at zero velocity until the muscle shortens isotonically, generating power at a constant point on the force-velocity curve. Finally, the muscle returns to the starting point of zero force and velocity when it relaxes. (B, left) This hypothetical work loop is modified when the contraction is repeated cyclically during locomotion. Any moving object (like an animal in motion) has kinetic energy. This energy can be captured as work done on the muscle to extend a muscle prior to shortening. Thus, while the muscle does the same (F-D) work on the environment, part of this work is derived from recovered strain energy stored in the muscle. The work loop is thus modified as the lengthening phase of the work loop is accomplished by the conversion of kinetic energy to work done on the muscle, as labeled, which reduces the work done by the muscle. (B, right) Once again, we can view this contraction as a power trajectory on the force-velocity plot. In this case, as the muscle is stretched (velocity is negative because it is in the opposite direction as the force produced) force increases until it reaches a magnitude sufficient for shortening to occur. From this point on, the power trajectory is similar to that seen above. (C, left) An *in vivo* work loop is indeed a loop that demonstrates characteristics not revealed in a force-velocity depiction. First, muscle fibers that have been lengthened during activation generate enhanced

(103). The utility of this approach is that the trajectory can be superimposed on the familiar Hill force-velocity curve itself. If muscle only shortens while contracting, and does so essentially isotonically, this trajectory would appear as a simple triangle as force increases at zero velocity quickly accelerates and is maintained at a maximum (and possibly sustained) power that approaches in magnitude, but never exceeds, the force-velocity curve, before returning to zero force and velocity. In this example (only illustrative) the power trajectory (Fig. 5A, right) is drawn to reflect the conventional wisdom that muscles operate at a shortening velocity about one-third maximum [$V_{\max}$: (86)] to maximize power output.

While work loops have been made for many animals, rarely do they appear as a rectangle. The exception is when these loops are made for locomotor muscles at very low cycle frequencies, substantially below *in vivo* use frequencies (102, 103). However, neither this “static” work loop nor power trajectory reflect those observed in animals during sustained locomotion. This begs the question “why not?” or perhaps more relevantly, what occurs at *in vivo* use frequencies that changes the shape of this work loop so that it no longer conforms to a Hill force-velocity model? Importantly, the recent evidence strongly suggests that force production and muscle shortening are often out of phase. Further, titin, the internal spring of muscle, likely plays an important role in this dynamic coupling (158). While this “Hill” work loop may fairly accurately predict the behavior of a muscle *in vitro* in a load clamp or the quadriceps *in vivo* while climbing a step, it fails to depict the work done by muscles in cyclic steady-state locomotion.

Whether we examine a fish swimming, an insect or bird in flight or a running mammal, sustained locomotion is characterized by cyclic alternations in muscle length resulting in recurring energy shifts as muscles store and recover potential energy. In the simplest case, the “negative work” absorbed by the muscle would reduce the area of the work loop by an amount reflecting the magnitude of the stored energy that can be successfully recovered. When the idealized work loop is modified to reflect this energy transfer, the area is greatly reduced, despite the fact that the shortening muscle is producing identical force for an identical distance (Fig. 5B, left). In other words, the animal is doing the same amount of work on the environment, but the muscle is doing much less positive work (and energetically costly; see Section “Energetics of Locomotion”). Some fraction of the elastic strain energy absorbed during muscle extension is recovered during shortening, minimizing muscle work. When we modify the expected power trajectory it now includes negative power as the curve encroaches into what Thomas MacMahon once labeled the

Figure 5 (Continued) force and thus work production. In addition, shortening is not isotonic, both force and velocity vary during shortening. (C, right panel) If we examine the power trajectory of this *in vivo* cyclic contraction, as force enhancement leads to increase in the work done by the muscle, the power produced during that part of the contraction similarly extends beyond the maximum power that a muscle can generate in an isolated (Hill) contraction.

“dark side” of the Hill force-velocity curve (132). However, one would predict that this curve should approach the force-velocity curve in the same way to reach the same maximum power output (Fig. 5B, right).

It is now apparent that when used cyclically, locomotor muscles deviate even further from the Hill model in two other respects. First, force produced during shortening is far from isotonic, but is initially highest and diminishes during the course of muscle shortening in insects (110), fish (6), birds (18), and mammals (8, 102). In addition, when muscle is activated for a period prior to shortening, the initial force it generates when it does shorten is increased over that predicted from static mechanics. Skeletal muscle has been found to produce substantially greater force if stretched (57, 83, 126) or even isometrically activated for some duration prior to shortening (123). A consequence of this muscle activation prior to shortening is that not only does muscle store and recover some of the elastic strain potential energy, but there is also a residual enhancement of force, above that predicted from the force-velocity relationship (103, 126) when it does shorten. This force enhancement, while transitory, results in substantially higher initial force and shortening velocity (and thus power) during muscle shortening (123). This transitory period of enhanced force and power will result in an increased amount of the total work done during a work cycle. However, the enhanced power is only available at a specific cycle frequencies (see below). When we incorporate these final two modifications into our idealized work loop, we see a resultant work loop that looks very similar to those described in a number of animals (Fig. 5C, left). Because force is amplified, the power trajectory described by this final idealized work loop will cross over the force-velocity curve. Animals utilizing this preactivation force enhancement can generate power that exceeds that predicted from the force-velocity curves generated with static properties (Fig. 5C, right).

We can easily put the work loop into context in a familiar model, running humans (Fig. 6). One complete locomotor cycle includes a stance phase, during which time the foot is in contact with the ground and a swing phase, when the foot is returned to prepare for the next stance (active) cycle. At footfall the knee extensors (quadriceps) are contracted and thus their shortest length. Until midstance when the center of mass is over the foot, the muscle is actively lengthened, as it absorbs work during deceleration. For the remainder of the stance phase the muscle shortens doing work until the foot leaves the ground and the muscle is again at its shortest length. During the swing phase, the muscle actually goes through a greater length excursion but does so producing minimal force.

Steady-state “cruising” locomotion

Most of the time animals are moving, the purpose seems to be simple displacement, that is, getting from point A to point B. Thus, most animals have an identifiable steady-state or “cruising” locomotion that is most often used while moving about. This cruising locomotion seems to follow the steady-state

properties of muscle discussed above. Because: (i) locomotor muscles must be stretched prior to each cycle and (ii) active stretch reduces muscle work and enhances force production, it would be advantageous to exploit these features of skeletal muscle during steady-state locomotion. Indeed, there is ample evidence that animals select cycle frequencies that maximize this property of muscle. Thus, in insects (109), fish (6, 201), birds (18), and mammals (102), there is a strong dependence of power output on cycle frequency, at frequencies too low (or too high) apparently much of the stored strain energy is dissipated as heat and thus no longer recoverable. As mentioned above, the work loops for these slow cycle frequencies appear close to rectangles (Fig. 5A, right). There is increasing evidence that the giant sarcomeric protein titin may play a key role in both energy storage and force enhancement (84, 149). Thus, in addition to the acknowledged role of tendons, there is a within muscle spring, the molecule titin. There is building evidence that this molecule itself is “tuned” to the frequency of muscle use, smaller stiffer titin isoforms are found in small (high muscle frequency) animals and larger, more compliant isoforms in large slower animals (134).

During sustained locomotion in all animals, muscles are used cyclically. Because activation and muscle length changes are “out of phase,” that is, often muscles are initially stretched while activated, this cyclic muscle use results in significant deviation from the “static” Hill force-velocity properties. In fact, locomotor muscles may typically have greater durations of activation while lengthening or contracting isometrically than while shortening and may even cease activation during the shortening phase (69, 123). Further, for any animal, there is an “optimal” frequency of cyclic muscle use that maximizes power output. This property of muscle is apparently inescapable, and is always linked to the cycle frequency in swimming, flying, and running animals.

Terrestrial locomotion

When animals move along the ground, they must exert forces to counteract gravity while generating propulsion. Thus, the resultant net ground forces reflect both body weight as well as forward thrust. Within a stride, ground forces include initial deceleration followed by larger propulsive forces. Further, as the speed of movement increases, quadrupeds go through a predictable succession of gait transitions from walking to trotting to galloping, reflecting different “Newtonian” mechanical solutions that minimize energy loss and thus maximize locomotor efficiency. At lower relative ground speeds, energy shifts primarily between kinetic and gravitational potential energy as the center of mass is elevated with each stride (125). As the animal moves forward, its center of mass falls slightly as the gravitational potential energy is converted back to kinetic energy (increasing slightly the forward speed). However, at higher speeds, the energy absorbed at footfall is stored as elastic recoil potential energy. It is this energy that may be either recovered in a subsequent stride or lost as heat, with the outcome primarily a function of cycle frequency (187).

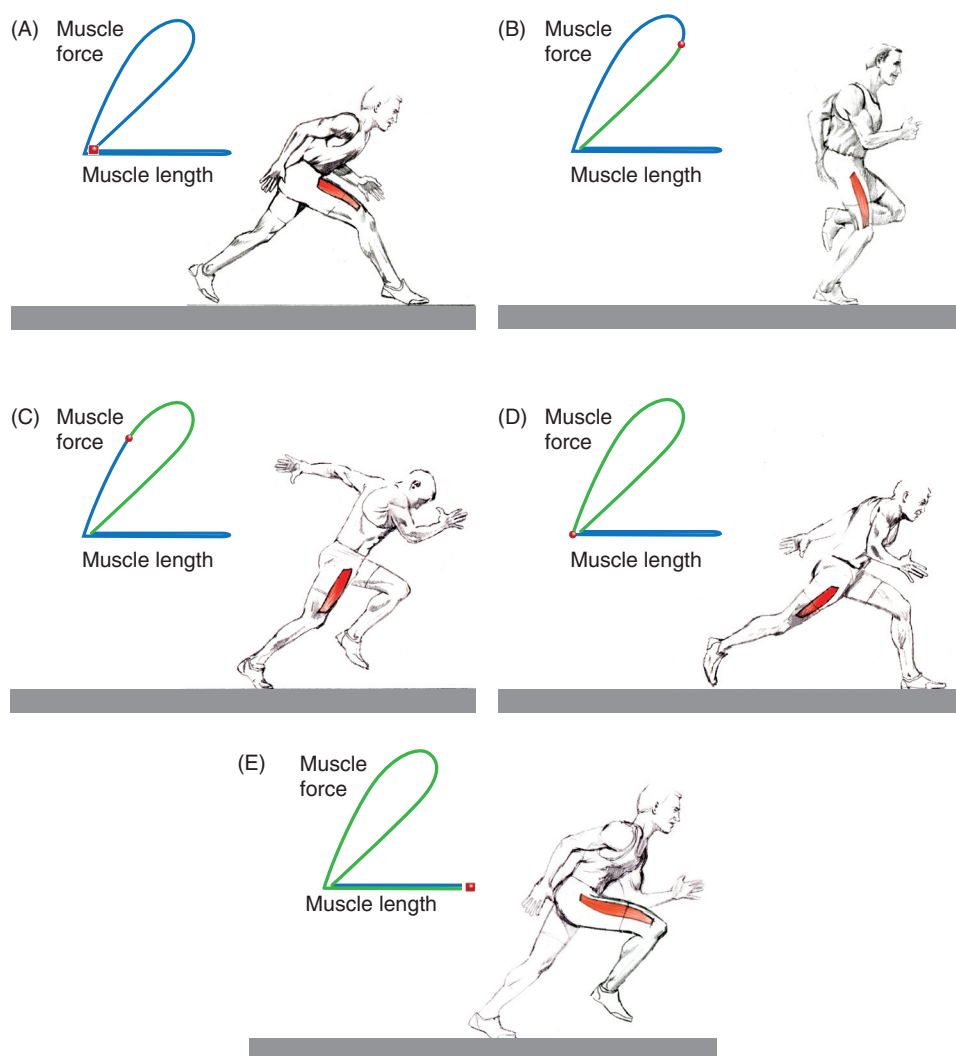


Figure 6 When animals run each stride represents an active and passive cycle of muscle lengthening and shortening. If we view one stride starting with foot-fall (A), the extensor muscles (quadriceps) are initially short as the knee is extended. During running, the center of mass remains relatively constant, thus the knee is slightly bent when beneath the center of mass (B) and thus has been actively lengthened in this portion of the stride, reflected in the accompanying work loop. The extensors are actively shortened until the foot leaves the ground (C and D) completing the active phase of the work loop. The recovery phase when the foot is off the ground is characterized by little work done but relatively longer muscle excursions, the flat phase of the work loop (E).

Because these patterns of locomotion transcend taxa (and extend to flying and swimming animals as well as discussed below), it is not unreasonable to suspect that locomotion is among the most fundamental influences constraining animal evolution [see reference (59)].

The natural experiment of allometry may be the best tool to demonstrate the importance of locomotor function as a constraint on other systems. Traditionally, allometric relations are simply described by the power law equation, $Y = aM^b$, where some variable Y can be predicted or described as a function of body mass (M). For a review of many of these relations, see references (35, 199). Among the emergent patterns revealed by allometry, none seems to be stronger than the relationship between stride frequency and body size. Body mass apparently sets stride frequency in running quadrupeds. A number

of investigators have derived this predicted scaling exponent and it ranges from $M^{-1/8}$ (147) to $M^{-1/6}$ (80). (Note, we use fractional exponents when discussing predictive relations and decimal exponents to report empirical data.) Stride frequency in running mammals generally scales with a body mass exponent of -0.13 to -0.21 (Table 2). At the physiologically equivalent (81) speed of the trot-gallop transition, mammals run with such predictable stride frequency ($r^2=0.84-0.98$) (Table 2) that galloping mammals can be modeled as simple mass-spring oscillators (125). It is notable that this dominant relationship transcends differences in athletic ability and even conflicting physiological demands. Two examples serve to illustrate this point.

First, among mammals of identical body size, there is often a wide range of aerobic capacity and running performance.

Perhaps none is more evident than the contrast among goats, dogs, and pronghorns. These similar sized animals vary in their maximum aerobic capacities and sustainable top speeds by factors of roughly 1:3:5. Not surprisingly, the specific cardiovascular and muscle properties responsible for the relative performance differences are profound (130, 184). However, despite the extreme divergence in running performance among these animals, doubtless the consequence of eons of unrelenting evolutionary pressure, galloping stride frequency of all three of these species is similar, conforming to the mass-spring model. In these animals and among quadrupeds in general, variable running speed is primarily a function of changes in stride length, not stride frequency. Running speed in pronghorns is five times that of running goats because they cover five times as much distance per stride; but similar sized goats and pronghorns gallop at the nearly identical stride frequency set by their similar body masses.

Second, we see that stride frequency takes precedence over other, even critical, physiological rates. Because of the importance of arterial pH, one of the most regulated of all physiological variables is ventilation, and hence the arterial partial pressure of carbon dioxide which directly sets arterial pH. The consequences of severe acidosis or alkalosis, including changes of protein structure, can be life threatening. In fact, the traditional textbook view of physiology is that arterial pH is the cornerstone of homeostasis.

In all running quadrupeds (at least above the size of a jackrabbit), there is a phase locking of limb and respiratory frequencies. In large mammals, the considerable mass of the digestive organs functions as an “abdominal piston” that repositions the diaphragm by momentum shifts with each stride as the running animal accelerates and decelerates but even in smaller animals the “respiratory muscles” participate in locomotor function (25). As a consequence, stride frequency and ventilation rate are inescapably linked in most running animals. Because of the critical importance of matching ventilation to CO₂ production, one might predict that stride frequency should be set to regulate minute volume during exercise, if possible. Perhaps this coupling is nowhere more dramatic than in thoroughbred horses. The body size-dependent stride frequency in galloping thoroughbreds results in a seemingly dangerous metabolism-ventilation mismatch. Their hypoventilation results in severe respiratory acidosis; arterial PCO₂ in running thoroughbreds has been measured as 56 mmHg (107). If running horses are given high CO₂ (6%) in the inspired air, the coupling remains though will switch from 1:1 to 2:1 ventilation:stride coupling (68). It seems that if stride frequency could be dissociated from the constraint of body size it certainly would be in these (and other large) animals. However, it is not.

The body size dependency of stride frequency is demonstrated in a physiology teaching laboratory that we use, inspired by the work of Claire Farley and C. R. Taylor. When hopping in place, a student volunteer quickly selects a preferred hopping frequency that feels “comfortable” and is highly reproducible for any given individual. By measuring

oxygen uptake, we can calculate the energetic cost per hop and thus efficiency. When forced to hop at half this frequency (controlling for height), the cost per hop doubles. The muscle spring operates with an apparent “resonant frequency” that maximizes the storage and recovery of elastic recoil potential energy. Further, when the specific preferred frequency is plotted on the graph that describes stride frequency as a function of body mass across a full size range of galloping mammals (Table 2), it falls very close to the value predicted for a mammal of the subject’s body size. Taylor has made the point that hopping on two feet in a biped is biomechanically similar to galloping in a quadruped, as demonstrated above. It is energetically costly to deviate from this preferred, energy-recovering frequency. During cyclic locomotion, muscles can recover 50% (or more) of the energy that would otherwise be lost by simply “tuning” the frequency of their use (132).

Because stride frequency is set to body size, all mechanical and energetic properties of the locomotor muscles animals of different body sizes might be expected to reflect this dominant constraint. Indeed, rates of muscle shortening in running mammals scale similarly to whole animal stride frequency (Table 2) and the frequency which provides the maximum power output in rabbit latissimus dorsi muscle obtained using *in vitro* work loops (5 Hz) is very close to the predicted stride frequency in a generic 1.5 kg mammal (4.3 Hz) (103), while the mouse Extensor Digitorum Longus produced maximum power output near 10 Hz, close to its predicted *in vivo* frequency of 8 Hz (102).

Movement through fluids: Swimming

Flight in birds and swimming in fish both require that propulsion be generated against a fluid that yields to propulsive forces, as opposed to the nonyielding terrestrial substrate. Much of our knowledge of the energetics and mechanics of propulsion in fish can be traced to the pioneering work of Paul Webb [see, e.g., reference (230)]. As is the case for terrestrial locomotion, fish have dramatic gait changes that occur with speed (115). This gait change is even more dramatic than that of running quadrupeds in that there is often a complete shift from pectoral fin (labriform) propulsion to body undulation (67). In addition, fin and tail beat frequencies in swimming are strongly body size dependent. The optimal undulation (tail-beat) frequency that maximizes power output is strongly linked to the body length of fish, F_{opt} (Hz) = $1.67 L \text{ cm}^{-0.52}$ (6). Because fish scale geometrically, mass varies as length cubed in fish (20), so the optimal tail-beat (undulation) frequency in fish must vary in proportion to $M^{-0.17}$, nearly identical to the body mass dependency of galloping in mammals.

When we examine how fish use their locomotor muscles, a pattern similar to terrestrial locomotion emerges. As is the case with running vertebrates, muscle activation in swimming fish during body undulation is initiated during the lengthening phase of cyclic muscle use. For example, in the red muscle of yellowfin tuna (a true aquatic athlete), the greatest duration of muscle activation (contraction) occurs while the muscle

is lengthening; interestingly most of the muscle shortening occurs when the muscle is no longer activated. As is the case in terrestrial locomotion, not only is power output dependent on cycle frequency, the optimal power output occurs only near *in vivo* cruising speed frequencies (201).

As fish are heterotherms, muscle power output is highly temperature dependent (5). The consequence of regional endothermy and elevated muscle temperature in those fish is elevated muscle power output. While only a small proportion of the total muscle mass, these red fibers produce “near maximal” mechanical power during normal cruise swimming (212).

While not the focus of this review, a great deal has been learned of the nature of fish propulsion in an aquatic medium using hydrodynamic analyses. For a review of vortex generation in fluid force production, see reference (124).

Movement through fluids: Flying

Flight presents additional challenges not present for most swimmers. Unlike most fish that are neutrally buoyant, birds must generate lift in addition to thrust. This task is complicated because of the extremely low viscosity of air relative to water. Nonetheless, we begin again with the identified similarities that birds share with fish and terrestrial vertebrates. Wings (airfoils) differ greatly in shape, and thus aspect ratio, along a continuum from short wide wings that favor maneuverability (flycatchers) to long flight-efficient, high aspect ratio wings (albatross). Despite these obvious differences, wing beat frequency is strongly body size dependent. Wing frequencies have been measured under various conditions and while there is still some apparent discrepancy among the measured values, the observed wing-beat frequencies for birds from 35 g to 8.5 kg scales as $M^{-0.17}$ (169). Note that once again the scaling exponent is nearly identical to locomotor frequencies measured in terrestrial locomotion and swimming fish. There has been substantial interest recently in exploring the mechanistic cause of this empirical relationship, leading to development of a theory of constant Strouhal numbers (222). This dimensionless number is the product of frequency and amplitude divided by speed. If animals maintain a constant Strouhal number wing beat frequency should vary between $M^{-1/6}$ and $M^{-1/3}$ (160).

While the center of mass does rise and fall with each wing beat cycle in most birds, these shifts in vertical position are minimized as the wings provide some lift on the upstroke as well as the down stroke of the wing. As the flight (air) speed of the bird increases and as the velocity of the wing upstroke decreases, lift is generated both as the wing moves down (providing forward thrust) and up, as the pectoral muscles return to their extended (lengthened) position (79). To maintain lift, the pectoralis muscles must maintain some low tension during the upstroke (thereby function as a brake slowing the upstroke). Indeed, force is developed during part of the upstroke, is maximized during the end of the upstroke and culminates in an isometric peak in tension prior to shortening. During

muscle shortening, force drops rapidly as the activation of the muscle stops very soon after peak tension is reached, and thus most of the shortening occurs after activation has ceased (18). In common with other vertebrates: first, cycle frequency is linked to body size and flight frequencies correspond to maxima of power outputs and second, muscle activation and muscle shortening are significantly out of phase.

Ballistic performance: Sprinting on land, in air, or water

Animals move for many reasons, most of which lack “urgency.” As the apparent “goal” of animal movement would most often seem to be a change in location, cost (energetic economy) of movement seems to have been the strongest selective pressure in sustained, “cruising,” locomotion (see Section “Energetics of Locomotion”) and thus animals apparently use their muscles to produce force in the most economical way.

However, survival occasionally depends on ballistic or burst movements that require peak power outputs for short durations to maximize speed of movement. Whether a cheetah running to capture an antelope, a frog leaping to escape predation, a fish in escape behavior, or the act of merely becoming airborne in a large bird, the success of these ballistic movements is coupled to (and thus doubtless selected by) maximizing instantaneous power output. This presents an interesting locomotor challenge in that usually the same muscles and skeletal elements responsible for sustained, economical movement must double as high-power generators.

Muscles, or muscle fibers, that best serve this function are fast contracting, with little cell volume devoted to mitochondria (see Fig. 1). It seems a safe assumption that, if ever maximum motor unit recruitment occurs, it is during these brief explosive events. Thus, while the same muscles are used for economical sustained movement and brief bursts, within the tissue the fibers powering these diverse tasks are usually distinct. In most animals from fish to Oryx, there is a large reservoir (the majority of the muscle mass) of fast contracting, (“white,” glycolytic) fibers that may be quiescent during sustained locomotion (or in some animals like modern humans, during most of the individual’s lifetime) that generate these high periodic power outputs (82). Most of the muscle mass of most animals is suited to, and reserved for, periodic, rare use. The overall muscle composition must therefore reflect two often opposing demands of burst and sustained locomotion, and thus there would seem to be a necessary trade-off between power and fatigue [(245) and Fig. 1].

In addition, these ballistic movements are characterized not only by high power output, but often also by a large magnitude in muscle shortening, beyond what would be permitted for the muscle to remain in the optimal portion of the length-tension relationship. In other words, when and where maximum power output may have the most critical survival consequences, muscle often shortens over lengths that sacrifice force because of their position on

the length-tension curve. However, it should be noted that as active force declines on this descending limb of the length-tension curve passive tension increases, so total force need not be sacrificed, as discussed below.

Sprinting on land

Running speed (and thus power output) varies in a hyperbolic pattern with duration of exercise. As can be seen from greyhound dogs, horses, or human running records, top average speed (from a standing start) occurs when running duration is about 10 to 20 s (108). Importantly, as stride lengths are maximized during sprinting, so are muscle length changes. It is not surprising that this is also thought to be the average duration of a cheetah's capture sprint, when they reach a top speed of 104 km h⁻¹ (29 m s⁻¹) (202). They are likely to give up the chase if they have no success after 20 s of sprinting after their prey (56). While this is simplest to conceptualize as dichotomous, in fact, power output (running speed) is a hyperbolic continuum of running duration, and in all instances, top speed is limited to very short, unsustainable, durations.

Frogs, fish, and birds: Burst performance

One aspect of burst muscle use that distinguishes it from sustained use is most obvious in frog jumping. The extent of muscle shortening during contraction is very high to generate the work required to jump; muscle strain is about 30% of muscle length (183). The problem thus arises at muscle lengths that are substantially off the length-tension plateau for active force production. The range of force predicted at these long muscle length changes could vary by 50%, obviously not a desirable outcome when maximum power is required. An important observation from Azizi and Roberts (10) is that the frog plantaris muscle starts at initially long lengths and shortens to optimal length. By starting at extended (stretched) muscle lengths, both the active and passive forces are available to these animals to power the jump.

To escape predation, fish use "fast-start" burst swimming that results in brief bouts of high swimming speeds by generating maximum muscle power. This behavior is also called a "C-start" as the animal begins this process by bending into a "C" shape. In addition to recruiting all of the white muscle fibers, this behavior also results in much greater muscle strain (stretch) than occurs in normal cruising locomotion. During the prestart bend, muscle strains of up to 20% have been recorded (227). Thus, as is the case with frog jumping, muscle length excursions extend well beyond the plateau of the active length-tension curve. Likewise, large birds in particular must achieve maximal power output for takeoff. It is during this behavior, when lift must be generated in absence of forward speed, that the entire mass of the flight muscles is engaged and wing excursions (and thus muscle strains) are maximized (18, 142). Thus, both birds and fish must maximize (short duration) use of all available muscle mass and do so with maximum muscle excursions.

Emergent principles

Running, flying, and swimming are biomechanically distinct and use very different muscle groups and skeletal elements. During cruising, all locomotor movements are cyclical and consequently two apparent principles emerge. First, muscle activation and muscle shortening are out of phase. Muscle activation precedes muscle shortening and often begins as the muscle is lengthening, providing, for the subsequent use cycle, both recoverable elastic strain energy and a boost in force enhancement. Second, across a range of body sizes, whether swimming, flying, or running, muscle cycle frequency is a fixed function of body size, in all instances scaling roughly as $M^{-1/6}$ (Table 2), a pattern that itself has spawned a broad range of hypotheses of movement itself (15). These dominant locomotor patterns emerge uniquely from the comparative approach; neither would be apparent in single species examinations. Unfortunately, identifying the patterns is only the first step in identifying the (more interesting) cause of this size-dependant frequency coupling. Many have speculated on this cause (30); however, the mechanisms remain unclear.

Ballistic, explosive bursts requiring high muscular power, are common to most vertebrates. While these usually employ the same muscles used in sustained locomotion, they often are accomplished by seldom used fast glycolytic fibers as the entire muscle mass is engaged in this maximal effort. In addition, these movements are usually characterized by the greatest muscle (hence sarcomere) length excursions. Thus, when animal maximize their speed, they do so by maximizing muscle recruitment as well as muscle strain, which necessarily requires muscle use at lengths on the descending limb of the active length-tension curve. This begs the question, when animals require the highest power output they apparently use their muscles at lengths that sacrifice force production. A possible solution to this riddle has been recently unraveled by Herzog and his colleagues. They have demonstrated that force is preserved in activated and stretched myofibrils, even if stretched to the point of minimal actin-myosin overlap (126). Apparently, this force preservation can be attributed to titin, which seems to function as a dynamic spring within muscle as it displays a calcium-dependent increase in stiffness (30, 84, 149, 158). In other words, when the muscles powering the running cheetah, jumping frog or fast-starting fish are stretched while activated, force is preserved, apparently even in the most extreme prestretching of the muscle prior to contraction.

Energetics of Locomotion

Metabolic characteristics of burst performance

Whenever an animal moves, there must be a supply of energy sufficient to meet the metabolic demand, whether as a short duration flood or as a sustained trickle. Ultimately, the capacity of muscle to generate power is a function of the mechanical machinery (3); however, there must be a linking of mechanical power output with metabolic power input. Although energy stores and delivery systems may have a high

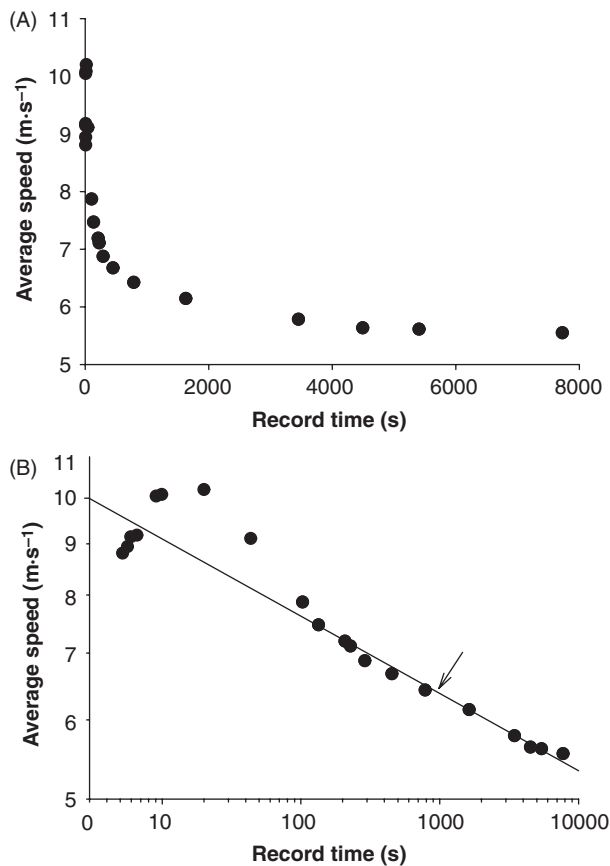


Figure 7 Maximum running speed in humans has been plotted using current world running records. (A) When these record times are plotted as a function of race duration, it seems to represent a dichotomous function; speed drops steeply over durations up to about 100 s and there is a nearly constant speed beyond that. (B) When the same data are plotted on logarithmic coordinates, different patterns emerge. First, because all races are from a standing start, the shortest duration events reflect this acceleration. Top speed is reached between 10 to 20 s. However, above 100 s, the relationship is linear ($r^2 = 0.98$). The arrow on this graph depicts the predicted speed (6.4 m s^{-1}) at a $\dot{V}_{O_2}$ max of $82 \text{ mL kg}^{-1} \text{ min}^{-1}$, corresponding to an elite runner.

capacity, they are not infinite: the high power outputs may be constrained by energy availability. Further, as the duration over which a muscle task must be performed is increased, the magnitude of metabolic power output that can be maintained decreases. This interaction of metabolic intensity and sustained power results in a strongly hyperbolic relationship (Fig. 7A); the maximum power output drops exponentially as the duration of exertion increases. As one might predict, the muscle structure of animals that rely on burst performance for their survival reflects this specialized adaptation; for example, burst power is maintained for only about 20 s in the cheetah, an organism with relatively low mitochondrial volume density and high anaerobic enzyme activity [Fig. 1; (244)].

The hyperbolic nature of the graph is generally interpreted as supporting a threshold effect such that anaerobic glycolysis is increasingly engaged when power requirements exceed the aerobic maximum ($\dot{V}_{O_2}$ max), but for decreasing durations. This paradigm suggests a digital response; an animal

can either use glycolysis for a short duration burst or oxidative phosphorylation sustainably. As mitochondrial ATP supply may be insufficient for performance, once an “anaerobic threshold” is exceeded, lactate accumulation ensues and performance duration is limited. It is worth noting that long-term sustainable metabolic activity cannot exceed about half of the locomotor muscles’ maximum aerobic capacity (78) with the possible exception of long-distance migration in birds (175).

In this context, metabolic power generation by skeletal muscle has been traditionally divided into two broad categories, each of which possesses defining characteristics that are related to the evolutionary forces that have shaped extant variation. The first, “burst power,” occurs when metabolic power is delivered at unsustainable rates for relatively short durations. In this mode, energy equivalents are made available via the creatine kinase reaction and glycolysis. Onboard phosphocreatine (PCr) can be instantly mobilized, while glycolysis rapidly yields accessible ATP, but at the price of lactate accumulation (141), and thus a defined limit to anaerobic stamina. Energy economy of use is of low importance when maximizing power output. Under these ballistic conditions, limitation to performance is found in the linking of the ability to generate mechanical power and in the availability of readily available onboard energy stores.

While the dichotomous nature of anaerobic glycolysis and aerobic performance has been a mainstay of biochemistry and introductory physiology texts for decades, a closer examination of the data suggests an analog, not digital response. First the idea that anaerobic glycolysis is the consequence of a lack of oxygen (conveniently suggested by the name) is likely never encountered save on the top of Everest. Indeed, there is now ample evidence that excess pyruvate, produced via glycolysis, is common even in prolonged activities (28, 114). It is apparent that this substrate is exported and used elsewhere, even during prolonged activity. This results in two locomotor consequences that are most obvious in viewing human running records (Fig. 7B). First, the record running speed that corresponds to an elite maximum oxygen uptake of $82 \text{ mL O}_2 (\text{kg} \cdot \text{min}^{-1})$, 6.25 m/s , is the speed maintained in a 6000 m run with a duration of 16 min, far longer a period than $\dot{V}_{O_2}$ max could be maintained. Second, the record for any race lasting from 100 s to over 2 h falls on an observed line with an r^2 of 0.976. There is an apparently exact relationship between activity duration and speed. It is only over very short durations that power output is high enough to cause this relationship to disappear.

Metabolic characteristics of prolonged performance

During sustained locomotion two factors related to muscle metabolism, rather than muscle mechanics, can conspire to limit performance. First, any disruption in the ionic composition, temperature, hydration state, or pH of the cell will increase the likelihood of development of muscle fatigue (60). Of particular relevance, a drop in pH inhibits ADP formation

and thus reduces the signal necessary for driving high rates of mitochondrial oxidative phosphorylation (42). Hence, homeostasis of the sarcoplasm is critical to allow the muscle fiber to function for long locomotor periods and influences the level of sustainable activity. Second, and the focus of sustained performance here, is the fact that energy supply must match energy demand.

Because animals use only a fraction of their available muscle mass for sustained locomotion, the steady-state rate of ATP synthesis, not muscle power capacity, must limit prolonged performance (42). The ATP synthetic rate is a complex function of many simultaneous variables, hence each of these can contribute to, or solely limit, sustained performance. The rates of oxygen supply, carbon dioxide and heat removal, and Krebs's cycle substrate delivery to the mitochondria, all contribute to the net rate of oxidative phosphorylation. Historically, there has been disproportionate attention given to just one of these, the rate of oxygen delivery; we begin our discussion here. The pathway for oxygen from the atmosphere to the muscle mitochondria is best depicted as a cascade of structures through which oxygen flows (234). Oxygen moves through each step of this cascade either by diffusion, with a cost of diminished partial pressure (lung to blood and blood to tissue), or by convection, the cost being a muscular pump (ventilation, cardiac output). The final oxygen sink, the skeletal muscle mitochondria will reflect the capacity and diffusive support (capillary density and blood flow) required to meet the steady-state ATP requirements.

In general, each step in the oxygen transport cascade varies in its ability to respond to demand within an organism's lifespan (phenotypic plasticity). The lung, confined in a rigid thoracic cavity, is perhaps the least malleable of these structures. As the lungs lack plasticity, they are thus "overbuilt" relative to demand, with excess structure allowing for a "reserve capacity." In contrast, the capillary and mitochondrial densities within skeletal muscle may be the most plastic and thus these structures closely reflect current demand (136). Perhaps intermediate, the cardiovascular system is also plastic, responding to an animal's changing aerobic requirements with structural modifications. While maximum heart rate is linked to body size alone (217) stroke volume can be increased. Thus, endurance training in many mammals has resulted in considerable cardiac hypertrophy; a training-induced doubling of maximum cardiac output is a direct consequence of the increase in stroke volume (189).

Perhaps because it is one of the simplest steps in the oxygen transport cascade to experimentally manipulate, cardiovascular oxygen delivery has historically received focused attention as a single rate-limiting step. Indeed, if oxygen delivery is experimentally limited (anemia, reduced cardiac output, hypoxia) oxygen uptake falls proportionately. This has been viewed by some as sufficient evidence to invoke a delivery limitation to oxygen uptake. However, if this step is supplemented, the additional oxygen delivery does not result in a proportional increase in oxygen consumption (129). Thus, rather than expecting a single rate-limiting step, we

might suggest a more reasonable hypothesis, that evolution over millennia has resulted in an oxygen transport system with "tuned" or equal capacities at each step of the cascade, just as we design assembly lines in factories. It would seem most reasonable that those structures with the greatest phenotypic plasticity should also be the structures likely to be just "sufficient" to meet demand. Thus, endurance training (in mammals) results in equal increases in oxygen delivery (increased hematocrit, cardiac output), the muscles' capacity for oxygen uptake (tissue capillarity) and use (mitochondrial densities within muscle). It should, therefore, come as no surprise that across taxa, there is a strong relationship ($r^2 = 0.88$) between capillary length density [J_V (c,f)] and mitochondrial volume density [V_V (mt,f)], such that J_V (c,f) = $258 + 1.25 V_V$ (mt,f). Using this correlation, Conley (41) calculated an average of 14 km of capillary per cubic centimeter of mitochondria. For review, see reference (136).

As the oxygen transport cascade terminates in mitochondria as the ultimate sink for oxygen, the role of oxygen demand in regulating mitochondrial structure and function is often stressed. In striated muscle, mitochondria are found in two pools; interfibrillar being in close proximity to the sarcomeres and subsarcolemmal just inside the cell membrane. It is frequently argued that subsarcolemmal mitochondria are so located to reduce the diffusion distance for oxygen. Implicit in this proposal is the assumption that movement of intracellular fuel substrates to the mitochondria or of ATP from the mitochondria to sites of use poses a lesser diffusion problem. We are not aware of direct measurements for diffusion of these compounds through the sarcoplasm, but indirect evidence suggests that an oxygen diffusion limitation is unlikely.

Measurements of intracellular PO_2 of working muscle using nuclear magnetic resonance have shown that PO_2 is preserved in working muscle (46). As intracellular oxygen levels are maintained as sufficiently high levels that oxidative phosphorylation is not inhibited, oxygen does not appear to play a regulatory role in the control of cellular respiration (140). That oxygen appears to be in sufficient supply within the muscle is supported as genetic ablation of myoglobin in mice led to no effect on exercise performance (65). That rats exposed to chronic hypoxia (45 days) showed no shift in mitochondrial localization (224) further argues against a diffusion limitation for oxygen leading to subsarcolemmal localization of mitochondria.

Were subsarcolemmal mitochondria accumulating in response to a problem in oxygen delivery, following the predictions of symmorphosis (236, 238), we would expect to see that smaller animals, with higher mass specific metabolic rates, would possess higher proportions of subsarcolemmal mitochondria. It has long been observed that greater volume densities of subsarcolemmal mitochondria accumulate in smaller mammals; however, muscles from these animals have higher total mitochondrial volume density, and thus possess larger volume densities of interfibrillar mitochondria as well. When the proportion of mitochondria found in the subsarcolemmal compartment is plotted against body size

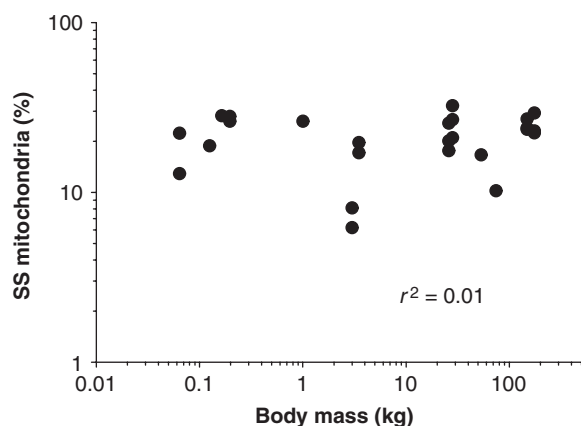


Figure 8 Mitochondria are localized in two compartments, subsarcolemmal and intermyofibrillar. It is commonly supposed that these compartments serve different functions, although these functions have not been identified. As smaller animals possess higher mass-specific metabolic rates, they also possess higher overall volume densities of mitochondria. However, the proportion of those mitochondria located in the subsarcolemmal compartment remains nearly constant across a wide range of body sizes. All data [from rat (33, 224), guinea pig (58), rabbit (180), cat (93), dog, goat, calf, and pony (94)] are from a variety of skeletal muscles, with variation in oxidative phenotype; however, when data from only the diaphragm are examined the outcome is nearly identical. Thus, increased metabolic demand is not met by increasing the proportion of mitochondria localized in the subsarcolemmal compartment.

(Fig. 8), the lack of any relationship further suggests that oxygen diffusion is not a factor in mitochondrial distribution. These data are collected from several different muscles, presumably with differing oxidative capacity, but if we include only those data from a single tissue, the diaphragm, for which a sufficient size range is available, the outcome is identical.

An alternative hypothesis for a subsarcolemmal localization of mitochondria is that they are so localized to minimize transport of fatty acids from the circulation. In those myocytes containing appreciable amounts of intracellular lipid, lipid droplets are always found in direct contact with mitochondria (225), suggesting that intracellular lipid transport is potentially limiting. In support of a role for fatty acid oxidation in subsarcolemmal localization, Koves et al. (118) found that subsarcolemmal mitochondria have higher respiration rates with lipid fuels than interfibrillar and respond to activity with a more robust enhancement of fatty acid oxidation.

In addition to oxygen, mitochondria need carbon sources to generate the reducing equivalents required for electron transport. For sustained muscle performance, oxygen delivery from the circulation must be equivalent to steady state demand as there is fairly negligible storage of oxygen in the muscle cell. Fuel supply must also be matched to demand; however, substrates for mitochondrial oxidation are primarily drawn from four sources, circulating lipids or glucose or stored intracellular lipids or glucose. All muscle cells store considerable glycogen and many also store large amounts of fatty acids in lipid droplets. Both are available for use during activity. Onboard fuels have the advantage of not requiring

import across the sarcolemma, but nonetheless must be mobilized. Glucose, stored as glycogen, is made available by the balance of glycogen phosphorylase and glycogen synthase activity (73). Intracellular (38) lipid droplets are found associated with mitochondria (225), presumably to minimize diffusion distance and fatty acids are transported across the mitochondrial outer membrane after binding with carnitine. Circulating fuels are acquired via sarcolemmal transporters. Uptake of glucose into muscle is dynamically regulated during activity such that import is greatly increased by sustained activity. The primary glucose transporter in muscle (GLUT 4) is mobilized from vesicles in the cytoplasm and inserted in the sarcolemma in response to activity through mechanisms that are not yet understood (38). While fatty acids are able to diffuse across the sarcolemma, specialized transporters ensure that lipid uptake is fairly constant during activity (231).

Sustained activity draws from these pools depending primarily upon the intensity of demand. As effort increases, the proportion of lipid fuel remains constant and glucose contributes an ever larger share of substrate (29, 184). In addition to changing substrates, the localization of the fuel source varies during activity. In a study of dogs and goats, animals with similar size, but different metabolic capacity, Taylor et al. (221) sought to test whether substrate delivery and use was tuned to demand. They found that as animals increased their locomotor demand, the contribution of extracellular glucose and fatty acids remained fairly constant while intracellular sources contributed a greater fraction with increased demand (232, 233). While the majority of the increased activity is fueled by the mobilization of glycogen (184), the increased lipid oxidative capacity of dogs (232) is possible due to greater lipid stores within the muscle cells (225). As intensity increases, the need to rely on intracellular sources suggests that capillary distribution is determined by oxygen rather than substrate demand (239) and that the capacity to store onboard fuels is critical during high intensity activity, both burst and sustained.

However, as activity is sustained for long durations, the importance of extracellular fuel sources may become more important. During a short burst activity, all of the fuel needs of the cell come from onboard sources, while in long-term, extreme marathon activity such as long-distance songbird migrations over open water, fuel needs must be met from extracellular sources. Understanding the constraints on extracellular uptake during higher intensity activities as described above thus also illuminates the metabolic ceiling that can be sustained over very long periods.

Economy of locomotion

Because oxidative phosphorylation requires intracellular organelles and extracellular oxygen (and often substrate), it is relatively slow and is limited by the total cristae surface area of mitochondria available. To reduce demand, economy (or efficiency) of energy transformation and execution of work is as important as increasing energy (ATP) availability. Examination of animal diversity demonstrates that behavior and

physiology often match this prediction; both cheetahs and dragsters possess engines that perform exceptionally well within the “burst” mode, fuel consumption does not seem to be a design criterion for either. However, as mentioned above, accumulation of waste products (heat, lactate) may be critical in ultimately limiting performance duration. In contrast, pronghorn antelope are exceptionally well adapted for economical sustained aerobic locomotor activity.

Mechanisms to maximize economy, or reduce the energetic cost of transport (COT; typically expressed as volume of oxygen consumed per gram body mass per meter moved or $\text{mLO}_2 \text{ g}^{-1} \text{ m}^{-1}$) thereby potentially act to facilitate sustainable locomotion. The pressure to reduce costs is likely most pronounced in those animals that operate at very high frequencies and thus have very high metabolic costs (45). Strategies to minimize demand include reduction of the work required for movement, for example, extremely light insect wings permit high-frequency, energy-intensive flight. Similarly, within clade comparisons indicate that species with more gracile limbs (muscle located more centrally) have the highest running speeds, at which energy costs of moving the limbs is highest (216). This observation is supported by experimental results from several species showing that weight added to the lower limb leads to a disproportional increase in energetic cost compared to weight added to the center of mass (154, 206, 241). A second strategy is to reduce force production itself, either structurally through lower myofilament volume density or functionally by lowering the number of cross-bridge cycles and thus muscle strain during each contraction. This strategy has reached its zenith in noisemaker muscles in which work requirements are very low (44) allowing for fiber volume to be usurped by mitochondria and SR (Fig. 1).

Although the energetic COT is a function of multiple physiological systems, it has long been observed that COT is higher in smaller than larger animals across taxa, whether running, swimming, or flying (198, 219). Thus, the well-described higher mass-specific metabolic capacities of smaller animals (237) are well matched to their sustained locomotor demands. As discussed previously, stride or wing and tail-beat frequency all appear to be strongly determined by body size (Table 2). Not surprisingly, as smaller animals operate at higher frequencies, not only do all muscles operate at higher frequencies, but their muscle tissue is also composed of a greater proportion of fast fiber types (77, 229). Fast fibers have higher energetic cost of force production and lower efficiency (207), likely contributing to the higher COT in these animals. Further, as the rates of locomotor cycles increase, the time that the limb is in contact with the ground during each cycle is reduced. Therefore, the rate of force production must be higher in smaller animals to support their mass against gravity, requiring a higher rate of energy consumption (119). These body size-driven relationships support that the relationship between COT and body size is the result of a constraint on locomotor performance driven by size-dependent operating frequency.

Either minimization of demand or enhancement of metabolic capacity are both plausible strategies for animals

to ensure that locomotion can be sustained. However, the vast majority of studies examining the ability of animals to match activity with sustainable locomotor performance have focused on muscle properties that increase mitochondrial capacity. There is ample evidence that skeletal muscle increases metabolic capacity to meet demand via muscle mitochondrial biogenesis (88, 90). There is some evidence that COT can be reduced, although this data is equivocal. Several studies on humans [reviewed in reference (49)] and rats (168) have not shown a difference in COT in comparisons of trained and untrained subjects. Similarly, a comparison of endurance and power trained human athletes (in which it was hypothesized that differences in muscle fiber types would be maximized) also showed no difference (121). While some studies have noted improvements in COT with training (151), the reduction in demand is only about 5%. The lack of difference in most studies is partly attributable to as yet unexplained, large interindividual variation. Variability within groups may reach 30%, even among highly trained athletes (49). In contrast to these data, the contractile cost of exercise (energy cost per unit force production) for humans performing isometric contractions while the rate of ATP use was measured with nuclear magnetic resonance spectroscopy was found to be significantly higher in human sprinters as compared to endurance trained distance runners (47), suggesting that variation at the muscle tissue may fail to produce energy savings during actual locomotion. Given its high metabolic capacity, skeletal muscle can be used for thermoregulation, in which economy is minimized to generate additional heat. One potentially unavoidable outcome of using muscles in this manner appears to be increased COT. Following chronic cold acclimation in which they relied upon muscle thermogenesis, COT was increased in short-tailed opossum (194). Several studies have used a comparative approach to running efficiency to evaluate variation in COT between four-legged and two-legged runners. In general, the COT for two-legged animals fits the inverse relationship between body size and COT described by Taylor et al. (219); however, humans appear to have unusually high COT, for reasons that remain unclear (215). A notable exception are large hopping mammals such as the kangaroo for whom the metabolic cost of locomotion is very high at low speeds, but remains unchanged or actually decreases with increasing speed (51). While there is considerable plasticity in muscle properties, with notable exceptions, the energetic COT appears to be constrained by size and frequency of use, although the mechanisms underlying these differences remain unclear.

Emergent principles

Locomotor activity can be constrained by numerous factors, mechanical and metabolic, depending upon the nature of the activity. The ability to deliver high-energy phosphate compounds (ATP, PCr) is an absolute requirement for force generation. However, evidence suggests that a dichotomy of aerobic versus anaerobic metabolism is an oversimplification. While

each mode of energy supply has distinct characteristics that are accentuated in the muscles of species or tissues specialized for different activities, it is apparent that both glycolysis and oxidative phosphorylation occur in parallel throughout many (if not most) locomotor activities. This basic principle is best illustrated by an extreme of muscle function, the rattlesnake tail-shaker muscle. Glycolysis provides nearly 40% of the ATP necessary for muscle work during sustained (up to hours) sound production, (114). As rattlesnake tail-shaker muscle makes up a relatively small percentage of body mass, the high lactate load can be transported away in the blood and consumed in other tissues. Aerobic production of lactate is not limited to sound production, but also occurs in humans, contributing nearly 20% of total ATP requirements (43) perhaps explaining why speeds exceeding $\dot{V}O_2$ max can be maintained for over 15 min in human runners. While lactate removal is less easily accomplished in a moving animal, it is commonly reported that the preferred fuel of the heart is lactate (28), a substance that would not be available were glycolysis only active during anaerobic conditions.

Reexamination of the relationship between locomotor speed and record times, examined logarithmically, reveals a regular, continuous increase in speed, and thus metabolic rate, that is achieved as the time that the activity can be sustained decreases. This suggests that throughout the range of locomotor performance, there is a regular contribution of both aerobic and anaerobic metabolism to ATP supply, whether above or below the anaerobic threshold (Fig. 7B). The consequence is that there is no “sustainable” aerobic performance, rather a continuum that quantitatively sacrifices power output for duration.

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References

- Abbott BC, Bigland B, Ritchie JM. The physiological cost of negative work. *J Physiol* 117: 380-390, 1952.
- Alexander Vernon. The dimensions of knee and ankle muscles and the forces they exert. *J Hum Movement Stud* 1: 115-123, 1975.
- Alexander RM. *Principles of Animal Locomotion*. Princeton, NJ; Oxford: Princeton University Press, 2003.
- Alexander RM, Jayes AS. Fourier analysis of forces exerted in walking and running. *J Biomech* 13: 383-390, 1980.
- Altringham JD, Block BA. Why do tuna maintain elevated slow muscle temperatures? Power output of muscle isolated from endothermic and ectothermic fish. *J Exp Biol* 200: 2617-2627, 1997.
- Altringham JD, Johnston IA. Scaling effects on muscle function: Power output of isolated fish muscle fibres performing oscillatory work. 1990.
- Alway SE, MacDougall JD, Sale DG, Sutton JR, McComas AJ. Functional and structural adaptations in skeletal muscle of trained athletes. *J Appl Physiol* 64: 1114-1120, 1988.
- Askew GN, Marsh RL. The effects of length trajectory on the mechanical power output of mouse skeletal muscles. *J Exp Biol* 200: 3119-3131, 1997.
- Azizi E, Brainerd EL, Roberts TJ. Variable gearing in pennate muscles. *Proc Natl Acad Sci U S A* 105: 1745-1750, 2008.
- Azizi E, Roberts TJ. Muscle performance during frog jumping: Influence of elasticity on muscle operating lengths. *Proc Biol Sci* 277: 1523-1530, 2010.
- Bandman E, Rosser BW. Evolutionary significance of myosin heavy chain heterogeneity in birds. *Microsc Res Tech* 50: 473-491, 2000.
- Barany M. ATPase activity of myosin correlated with speed of muscle shortening. *J Gen Physiol* 50: (Suppl): 197-218, 1967.
- Barclay CJ, Woledge RC, Curtin NA. Energy turnover for Ca²⁺ cycling in skeletal muscle. *J Muscle Res Cell Motil* 28: 259-274, 2007.
- Bassel-Duby R, Olson EN. Signaling pathways in skeletal muscle remodeling. *Annu Rev Biochem* 75: 19-37, 2006.
- Bejan A, Marden JH. Unifying constructal theory for scale effects in running, swimming and flying. *J Exp Biol* 209: 238-248, 2006.
- Bennett MJ, Rinaldo P, Strauss AW. Inborn errors of mitochondrial fatty acid oxidation. *Crit Rev Clin Lab Sci* 37: 1-44, 2000.
- Bicudo JE, Zerbini CV. Physiological constraints in the aerobic performance of hummingbirds. *Braz J Med Biol Res* 28: 1139-1145, 1995.
- Biewener AA, Corning WR, Tobalske BW. In vivo pectoralis muscle force-length behavior during level flight in pigeons (*Columba livia*). *J Exp Biol* 201 (Pt 24): 3293-307, 1998.
- Biewener AA, Roberts TJ. Muscle and tendon contributions to force, work, and elastic energy savings: A comparative perspective. *Exerc Sport Sci Rev* 28: 99-107, 2000.
- Bister TJ, Willis DW, Brown ML, Jordan SM, Neumann RM, Quist MC, Guy CS. Proposed standard weight (W_s) equations and standard length categories for 18 warmwater nongame and riverine fish species. *N Am J Fish Manage* 20: 570-574, 2000.
- Block BA. Thermogenesis in muscle. *Annu Rev Physiol* 56: 535-577, 1994.
- Bodine SC, Roy RR, Eldred E, Edgerton VR. Maximal force as a function of anatomical features of motor units in the cat tibialis anterior. *J Neurophysiol* 57: 1730-1745, 1987.
- Boggs DF, Frappell PB. Unifying principles of locomotion: Foreword. *Physiol Biochem Zool* 73: 647-650, 2000.
- Bottinelli R, Reggiani C. Human skeletal muscle fibres: Molecular and functional diversity. *Prog Biophys Mol Biol* 73: 195-262, 2000.
- Bramble DM, Carrier DR. Running and breathing in mammals. *Science* 219: 251-6, 1983.
- Brini M, Carafoli E. Calcium pumps in health and disease. *Physiol Rev* 89: 1341-1378, 2009.
- Brooke MH, Kaiser KK. Three human myosin ATPase systems and their importance in muscle pathology. *Neurology* 20: 404-5, 1970.
- Brooks GA. Cell-cell and intracellular lactate shuttles. *J Physiol* 587: 5591-600, 2009.
- Brooks GA, Mercier J. Balance of carbohydrate and lipid utilization during exercise: The “crossover” concept. *J Appl Physiol* 76: 2253-2261, 1994.
- Brown JH, West GB, Santa Fe Institute (Santa Fe N.M.). *Scaling in Biology*. Oxford; New York: Oxford University Press, 2000.
- Bruderer B, Peter D, Boldt A, Liechti F. Wing-beat characteristics of birds recorded with tracking radar and cine camera. *IBIS* 152: 272-291, 2010.
- Buller AJ, Eccles JC, Eccles RM. Differentiation of fast and slow muscles in the cat hind limb. *J Physiol* 150: 399-416, 1960.
- Buser KS, Kopp B, Gehr P, Weibel ER, Hoppeler H. Effect of cold environment on skeletal muscle mitochondria in growing rats. *Cell Tissue Res* 225: 427-436, 1982.
- Caiozzo VJ. Plasticity of skeletal muscle phenotype: Mechanical consequences. *Muscle Nerve* 26: 740-768, 2002.
- Calder WA. *Size, Function, and Life History*. London: Harvard University Press, Cambridge, Massachusetts, 1984, p. 431.
- Canto C, Auwerx J. AMP-activated protein kinase and its downstream transcriptional pathways. *Cell Mol Life Sci* 67: 3407-3423, 2010.
- Carrier DR, Heglund NC, Earls KD. Variable gearing during locomotion in the human musculoskeletal system. *Science* 265: 651-3, 1994.
- Cartee GD, Wojtaszewski JF. Role of Akt substrate of 160 kDa in insulin-stimulated and contraction-stimulated glucose transport. *Appl Physiol Nutr Metab* 32: 557-566, 2007.
- Cediel RA, Blob RW, Schrank GD, Plourde RC, Schoenfuß HL. Muscle fiber type distribution in climbing Hawaiian gobioid fishes: Ontogeny and correlations with locomotor performance. *Zoology (Jena)* 111: 114-122, 2008.

40. Chin ER, Olson EN, Richardson JA, Yang Q, Humphries C, Shelton JM, Wu H, Zhu W, Bassel-Duby R, Williams RS. A calcineurin-dependent transcriptional pathway controls skeletal muscle fiber type. *Genes Dev* 12: 2499-509, 1998.
41. Conley KE, Kayar SR, Rosler K, Hoppeler H, Weibel ER, Taylor CR. Adaptive variation in the mammalian respiratory system in relation to energetic demand. 4. Capillaries and their relationship to oxidative capacity. *Respir Physiol* 69: 47-64, 1987.
42. Conley KE, Kemper WF, Crowther GJ. Limits to sustainable muscle performance: Interaction between glycolysis and oxidative phosphorylation. *J Exp Biol* 204: 3189-3194, 2001.
43. Conley KE, Kushmerick MJ, Jubrias SA. Glycolysis is independent of oxygenation state in stimulated human skeletal muscle in vivo. *J Physiol* 511 (Pt 3): 935-945, 1998.
44. Conley KE, Lindstedt SL. Minimal cost per twitch in rattlesnake tail muscle. *Nature* 383: 71-72, 1996.
45. Conley KE, Lindstedt SL. Energy-saving mechanisms in muscle: The minimization strategy. *J Exp Biol* 205: 2175-2181, 2002.
46. Conley KE, Ordway GA, Richardson RS. Deciphering the mysteries of myoglobin in striated muscle. *Acta Physiologica Scandinavica* 168: 623-634, 2000.
47. Crowther GJ, Jubrias SA, Gronka RK, Conley KE. A "functional biopsy" of muscle properties in sprinters and distance runners. *Med Sci Sports Exerc* 34: 1719-1724, 2002.
48. Daniel TL, Tu MS. Animal movement, mechanical tuning and coupled systems. *J Exp Biol* 202: 3415-3421, 1999.
49. Daniels JT. A physiologist's view of running economy. *Med Sci Sports Exerc* 17: 332-8, 1985.
50. Darveau CA, Hochachka PW, Welch KC, Roubik DW, Suarez RK. Allometric scaling of flight energetics in Panamanian orchid bees: A comparative phylogenetic approach. *J Exp Biol* 208: 3581-3591, 2005.
51. Dawson TJT, Richard C. Energetic cost of locomotion in kangaroos. *Nature* 246: 313-314, 1973.
52. Dellung U, Tureckova J, Lim HW, De Windt LJ, Rotwein P, Molkentin JD. A calcineurin-NFATc3-dependent pathway regulates skeletal muscle differentiation and slow myosin heavy-chain expression. *Mol Cell Biol* 20: 6600-6611, 2000.
53. Desplanches D, Kayar SR, Sempore B, Flandrois R, Hoppeler H. Rat soleus muscle ultrastructure after hindlimb suspension. *J Appl Physiol* 69: 504-508, 1990.
54. Dickinson MH, Farley CT, Full RJ, Koehl MA, Kram R, Lehman S. How animals move: An integrative view. *Science* 288: 100-6, 2000.
55. Drucker E, Jensen J. Pectoral fin locomotion in the striped surfperch. II. Scaling swimming kinematics and performance at a gait transition. *J Exp Biol* 199: 2243-2252, 1996.
56. Eaton RL. *Cheetah: The Biology, Ecology, and Behavior of an Endangered Species*. New York: Van Nostrand Reinhold Co., 1974.
57. Edman KA, Elzinga G, Noble MI. Further characterization of the enhancement of force by stretch during activity in single muscle fibres of the frog [proceedings]. *J Physiol* 280: 35P-36P, 1978.
58. Eisenberg BR, Kuda AM, Peter JB. Stereological analysis of mammalian skeletal muscle. I. Soleus muscle of the adult guinea pig. *J Cell Biol* 60: 732-754, 1974.
59. Feder ME, Garland T, Jr, Marden JH, Zera AJ. Locomotion in response to shifting climate zones: Not so fast. *Annu Rev Physiol* 72: 167-190, 2010.
60. Fitts RH. Muscle fatigue: The cellular aspects. *Am J Sports Med* 24: S9-S13, 1996.
61. Flann KL, LaStayo PC, McClain DA, Hazel M, Lindstedt SL. Muscle damage and muscle remodeling: No pain, no gain? *J Exp Biol* 214: 674-679, 2011.
62. Folland JP, Williams AG. The adaptations to strength training: Morphological and neurological contributions to increased strength. *Sports Med* 37: 145-168, 2007.
63. Francis-West PH, Antoni L, Anakwe K. Regulation of myogenic differentiation in the developing limb bud. *J Anat* 202: 69-81, 2003.
64. Gardiner KR, Gardiner PF, Edgerton VR. Guinea pig soleus and gastrocnemius electromyograms at varying speeds, grades, and loads. *J Appl Physiol* 52: 451-7, 1982.
65. Garry DJ, Ordway GA, Lorenz JN, Radford NB, Chin ER, Grange RW, Bassel-Duby R, Williams RS. Mice without myoglobin. *Nature* 395: 905-908, 1998.
66. Gatesy SM, Biewener AA. Bipedal locomotion - effects of speed, size and limb posture in birds and humans. *J Zool* 224: 127-147, 1991.
67. Gibb A, Jayne B, Lauder G. Kinematics of pectoral fin locomotion in the bluegill sunfish *lepisomis macrochirus*. *J Exp Biol* 189: 133-161, 1994.
68. Gillespie JR, Landgren GL, Leith DE. 1:2 ratio of breathing to stride frequencies in a galloping horse breathing 6% CO₂. *Equine Exercise Physiology* 3: 66-70, 1991.
69. Gillis GB, Blob RW. How muscles accommodate movement in different physical environments: Aquatic vs. terrestrial locomotion in vertebrates. *Comp Biochem Physiol A Mol Integr Physiol* 131: 61-75, 2001.
70. Goldspink G. Gene expression in skeletal muscle. *Biochem Soc Trans* 30: 285-290, 2002.
71. Goldspink G. Research on mechano growth factor: Its potential for optimising physical training as well as misuse in doping. *Br J Sports Med* 39: 787-788; discussion 787-788, 2005.
72. Goldspink G, Scutt A, Martindale J, Jaenicke T, Turay L, Gerlach GF. Stretch and force generation induce rapid hypertrophy and myosin isoform gene switching in adult skeletal muscle. *Biochem Soc Trans* 19: 368-373, 1991.
73. Greenberg CC, Jurczak MJ, Danos AM, Brady MJ. Glycogen branches out: New perspectives on the role of glycogen metabolism in the integration of metabolic pathways. *Am J Physiol Endocrinol Metab* 291: E1-E8, 2006.
74. Gregorio CC, Granzier H, Sorimachi H, Labeit S. Muscle assembly: A titanic achievement? *Curr Opin Cell Biol* 11: 18-25, 1999.
75. Guyton AC. *Human Physiology and Mechanisms of Disease*. Philadelphia, Saunders, 1987.
76. Haldane JB. On being the right size. In: Possible Worlds and Other Pappus. New York: Harper & Brothers, 20-28, 1928.
77. Hamalainen N, Pette D. Patterns of myosin isoforms in mammalian skeletal muscle fibres. *Microsc Res Tech* 30: 381-389, 1995.
78. Hammond KA, Diamond J. Maximal sustained energy budgets in humans and animals. *Nature* 386: 457-462, 1997.
79. Hedrick TL, Usherwood JR, Biewener AA. Wing inertia and whole-body acceleration: An analysis of instantaneous aerodynamic force production in cockatiels (*Nymphicus hollandicus*) flying across a range of speeds. *J Exp Biol* 207: 1689-702, 2004.
80. Heglund NC, Taylor CR. Speed, stride frequency and energy cost per stride: How do they change with body size and gait? *J Exp Biol* 138: 301-318, 1988.
81. Heglund NC, Taylor CR, McMahon TA. Scaling stride frequency and gait to animal size: Mice to horses. *Science* 186: 1112-3, 1974.
82. Henneman E. The size-principle: A deterministic output emerges from a set of probabilistic connections. *J Exp Biol* 115: 105-112, 1985.
83. Herzog W, Leonard TR. Force enhancement following stretching of skeletal muscle: A new mechanism. *J Exp Biol* 205: 1275-1283, 2002.
84. Herzog W, Leonard TR, Joumaa V, Mehta A. Mysteries of muscle contraction. *J Appl Biomech* 24: 1-13, 2008.
85. Hill AV. The heat of shortening and the dynamic constants of muscle. *Proc R Soc Lond B* 126: 136-195, 1938.
86. Hill AV. The series elastic component of muscle. *Proc R Soc Lond B Biol Sci* 137: 273-280, 1950.
87. Hochachka PW, Somero GN. *Biochemical Adaptation: Mechanism and Process in Physiological Evolution*. Oxford: Oxford University Press, 2002.
88. Holloszy JO. Regulation of mitochondrial biogenesis and GLUT4 expression by exercise. *Compr Physiol* 1: 921-940, 2011.
89. Hood DA, Ircher I, Ljubicic V, Joseph AM. Coordination of metabolic plasticity in skeletal muscle. *J Exp Biol* 209: 2265-2275, 2006.
90. Hood DA, Uguccioni G, Vainshtein A, D'souza D. Mechanisms of exercise induced mitochondrial biogenesis in skeletal muscle: Implications for health and disease. *Compr Physiol* 1: 1119-1134, 2011.
91. Hoppeler H, Baum O, Lurman G, Mueller M. Molecular mechanisms of muscle plasticity with exercise. *Compr Physiol* 1: 1383-1412, 2011.
92. Hoppeler H, Howald H, Conley K, Lindstedt SL, Claassen H, Vock P, Weibel ER. Endurance training in humans: Aerobic capacity and structure of skeletal muscle. *J Appl Physiol* 59: 320-7, 1985.
93. Hoppeler H, Hudlicka O, Uhlmann E. Relationship between mitochondria and oxygen consumption in isolated cat muscles. *J Physiol* 385: 661-675, 1987a.
94. Hoppeler H, Kayar SR, Claassen H, Uhlmann E, Karas RH. Adaptive variation in the mammalian respiratory system in relation to energetic demand: III. Skeletal muscles: setting the demand for oxygen. *Resp Physiol* 69: 27-46, 1987b.
95. Hoppeler H, Luthi P, Claassen H, Weibel ER, Howald H. The ultrastructure of the normal human skeletal muscle. A morphometric analysis on untrained men, women and well-trained orienteers. *Pflügers Arch* 344: 217-232, 1973.
96. Hou TT, Johnson JD, Rall JA. Parvalbumin content and Ca²⁺ and Mg²⁺ dissociation rates correlated with changes in relaxation rate of frog muscle fibres. *J Physiol* 441: 285-304, 1991.
97. Hu P, Zhang KM, Spratt JA, Wechsler AS, Briggs FN. Transcription rates of SERCA and phospholamban genes change in response to chronic stimulation of skeletal muscle. *Biochim Biophys Acta* 1395: 121-125, 1998.
98. Hudson NJ, Franklin CE. Maintaining muscle mass during extended disuse: Aestivating frogs as a model species. *J Exp Biol* 205: 2297-303, 2002.
99. Hulbert AJ, Else PL. Membranes and the setting of energy demand. *J Exp Biol* 208: 1593-9, 2005.
100. Hurley BF, Redmond RA, Pratley RE, Truth MS, Rogers MA, Goldberg AP. Effects of strength training on muscle hypertrophy and muscle cell disruption in older men. *Int J Sports Med* 16: 378-384, 1995.

101. Jackman MR, Willis WT. Characteristics of mitochondria isolated from type I and type IIb skeletal muscle. *Am J Physiol* 270: C673-C678, 1996.
102. James RS, Altringham JD, Goldspink DF. The mechanical properties of fast and slow skeletal muscles of the mouse in relation to their locomotory function. *J Exp Biol* 198: 491-502, 1995.
103. James RS, Young IS, Cox VM, Goldspink DF, Altringham JD. Isometric and isotonic muscle properties as determinants of work loop power output. *Pflügers Arch* 432: 767-774, 1996.
104. Jansson E, Kaijser L. Effect of diet on muscle glycogen and blood glucose utilization during a short-term exercise in man. *Acta Physiol Scand* 115: 341-347, 1982.
105. Johnston IA, Temple GK. Thermal plasticity of skeletal muscle phenotype in ectothermic vertebrates and its significance for locomotory behaviour. *J Exp Biol* 205: 2305-2322, 2002.
106. Johnston II, Calvo J, Guderley YH. Latitudinal variation in the abundance and oxidative capacities of muscle mitochondria in perciform fishes. *J Exp Biol* 201 (Pt 1): 1-12, 1998.
107. Jones JH. Optimization of the mammalian respiratory system: Symmorphosis versus single species adaptation. *Comp Biochem Physiol B Biochem Mol Biol* 120: 125-138, 1998.
108. Jones JH, Lindstedt SL. Limits to maximal performance. *Annu Rev Physiol* 55: 547-569, 1993.
109. Josephson RK, Stevenson RD. The efficiency of a flight muscle from the locust *Schistocerca americana*. *J Physiol* 442: 413-429, 1991.
110. Josephson RK, Young D. Synchronous and asynchronous muscles in cicadas. *J Exp Biol* 91: 219-237, 1981.
111. Josephson RK, Young D. A synchronous insect muscle with an operating frequency greater than 500 Hz. *J Exp Biol* 118: 185-208, 1985.
112. Kardon G, Campbell JK, Tabin CJ. Local extrinsic signals determine muscle and endothelial cell fate and patterning in the vertebrate limb. *Dev Cell* 3: 533-545, 2002.
113. Kelly DP, Scarpulla RC. Transcriptional regulatory circuits controlling mitochondrial biogenesis and function. *Genes Dev* 18: 357-368, 2004.
114. Kemper WF, Lindstedt SL, Hartzler LK, Hicks JW, Conley KE. Shaking up glycolysis: Sustained, high lactate flux during aerobic rattling. *Proc Natl Acad Sci U S A* 98: 723-728, 2001.
115. Kendall JL, Lucey KS, Jones EA, Wang J, Ellerby DJ. Mechanical and energetic factors underlying gait transitions in bluegill sunfish (*Lepomis macrochirus*). *J Exp Biol* 210: 4265-4271, 2007.
116. Kleiber M. Body size and metabolism. *Physiol* 27: 511-541, 1932.
117. Kokshenev VB. Key principle of the efficient running, swimming, and flying. *Europhysics Letters* 90: 1-5, 2010.
118. Koves TR, Noland RC, Bates AL, Henes ST, Muoio DM, Cortright RN. Subsarcolemmal and intermyofibrillar mitochondria play distinct roles in regulating skeletal muscle fatty acid metabolism. *Am J Physiol Cell Physiol* 288: C1074-C1082, 2005.
119. Kram R, Taylor CR. Energetics of running: A new perspective. *Nature* 346: 265-267, 1990.
120. Krogh A. The progress of physiology. *Science* 70: 200-204, 1929.
121. Kyrolainen H, Komi PV. Differences in mechanical efficiency between power- and endurance-trained athletes while jumping. *Eur J Appl Physiol Occup Physiol* 70: 36-44, 1995.
122. Labeit D, Watanabe K, Witt C, Fujita H, Wu Y, Lahmers S, Funck T, Labeit S, Granzier H. Calcium-dependent molecular spring elements in the giant protein titin. *Proc Natl Acad Sci U S A* 100: 13716-13721, 2003.
123. Lappin AK, Monroy JA, Pilarski JQ, Zepnewski ED, Pierotti DJ, Nishikawa KC. Storage and recovery of elastic potential energy powers ballistic prey capture in toads. *J Exp Biol* 209: 2535-2553, 2006.
124. Lauder GV, Drucker EG. Forces, fishes, and fluids: Hydrodynamic mechanisms of aquatic locomotion. *News Physiol Sci* 17: 235-240, 2002.
125. Lee CR, Farley CT. Determinants of the center of mass trajectory in human walking and running. *J Exp Biol* 201: 2935-2944, 1998.
126. Leonard TR, Herzog W. Regulation of muscle force in the absence of actin-myosin-based cross-bridge interaction. *Am J Physiol Cell Physiol* 299: C14-C20, 2010.
127. LeWinter MM, Granzier H. Cardiac titin: A multifunctional giant. *Circulation* 121: 2137-2145, 2010.
128. Lewontin RC. *The Triple Helix: Gene, Organism, and Environment*. London: Harvard University Press, Cambridge, Mass, 2000, p. 136.
129. Lindstedt SL, Conley KE. Human aerobic performance: Too much ado about limits to V(O₂). *J Exp Biol* 204: 3195-3199, 2001.
130. Lindstedt SL, Hokanson JF, Wells DJ, Swain SD, Hoppeler H, Navarro V. Running energetics in the pronghorn antelope. *Nature* 353: 748-750, 1991.
131. Lindstedt SL, Hoppeler H, Bard KM, Thronson HA, Jr. Estimate of muscle-shortening rate during locomotion. *Am J Physiol* 249: R699-R703, 1985.
132. Lindstedt SL, LaStayo PC, Reich TE. When active muscles lengthen: Properties and consequences of eccentric contractions. *News Physiol Sci* 16: 256-261, 2001.
133. Lindstedt SL, McGlothlin T, Percy E, Pifer J. Task-specific design of skeletal muscle: Balancing muscle structural composition. *Comp Biochem Physiol B Biochem Mol Biol* 120: 35-40, 1998.
134. Lindstedt SL, Reich TE, Keim P, LaStayo PC. Do muscles function as adaptable locomotor springs? *J Exp Biol* 205: 2211-6, 2002.
135. Lindstedt SL, Thomas RG, Leith DE. Does peak inspiratory flow contribute to setting VO₂max? A test of symmorphosis. *Respir Physiol* 95: 109-118, 1994.
136. Lindstedt SL, Wells DJ, Jones JH, Hoppeler H, Thronson HA, Jr. Limitations to aerobic performance in mammals: Interaction of structure and demand. *Int J Sports Med* 9: 210-217, 1988.
137. Luo Y, Davis JP, Tikunova SB, Smillie LB, Rall JA. Myofibrillar determinants of rate of relaxation in skinned skeletal muscle fibers. *Adv Exp Med Biol* 538: 573-581; discussion 581-582, 2003.
138. Luthi JM, Howald H, Claassen H, Rosler K, Vock P, Hoppeler H. Structural changes in skeletal muscle tissue with heavy-resistance exercise. *Int J Sports Med* 7: 123-127, 1986.
139. MacDougall JD, Sale DG, Moroz JR, Elder GC, Sutton JR, Howald H. Mitochondrial volume density in human skeletal muscle following heavy resistance training. *Med Sci Sports* 11: 164-6, 1979.
140. Marcinek DJ, Ciesielski WA, Conley KE, Schenkan KA. Oxygen regulation and limitation to cellular respiration in mouse skeletal muscle in vivo. *Am J Physiol Heart Circ Physiol* 285: H1900-H1908, 2003.
141. Marcinek DJ, Kushmerick MJ, Conley KE. Lactic acidosis in vivo: Testing the link between lactate generation and H⁺ accumulation in ischemic mouse muscle. *J Appl Physiol* 108: 1479-1486, 2010.
142. Marden JH. Variability in the size, composition, and function of insect flight muscles. *Annu Rev Physiol* 62: 157-178, 2000.
143. Marsh RL. How muscles deal with real-world loads: The influence of length trajectory on muscle performance. *J Exp Biol* 202: 3377-3385, 1999.
144. Maruyama K, Natori R, Nonomura Y. New elastic protein from muscle. *Nature* 262: 58-60, 1976.
145. Mathieu O, Krauer R, Hoppeler H, Gehr P, Lindstedt SL, Alexander RM, Taylor CR, Weibel ER. Design of the mammalian respiratory system. VII. Scaling mitochondrial volume in skeletal muscle to body mass. *Respir Physiol* 44: 113-128, 1981.
146. Matsakas A, Patel K. Skeletal muscle fibre plasticity in response to selected environmental and physiological stimuli. *Histol Histopathol* 24: 611-629, 2009.
147. McMahon TA. Using body size to understand the structural design of animals: Quadrupedal locomotion. *J Appl Physiol* 39: 619-627, 1975.
148. Mendez J, Keys A. Density and composition of mammalian muscle. *Metabolism* 9: 184-188, 1960.
149. Monroy JA, Powers KL, Gilmore LA, Uyeno TA, Lindstedt SL, Nishikawa KC. What is the role of titin in active muscle? *Exerc Sport Sci Rev* 40: 73-78, 2012.
150. Moon BR, Conley KE, Lindstedt SL, Urquhart MR. Minimal shortening in a high-frequency muscle. *J Exp Biol* 206: 1291-1297, 2003.
151. Morgan DW, Bransford DR, Costill DL, Daniels JT, Howley ET, Krahenbuhl GS. Variation in the aerobic demand of running among trained and untrained subjects. *Med Sci Sports Exerc* 27: 404-409, 1995.
152. Mounier Y, Tiffreau V, Montel V, Bastide B, Stevens L. Phenotypical transitions and Ca²⁺ activation properties in human muscle fibers: Effects of a 60-day bed rest and countermeasures. *J Appl Physiol* 106: 1086-1099, 2009.
153. Mountcastle AM, Daniel TL. Vortexlet models of flapping flexible wings show tuning for force production and control. *Bioinspiration & Biomimetics* 5: 045005, 2010.
154. Myers MJ, Steudel K. Effect of limb mass and its distribution on the energetic cost of running. *J Exp Biol* 116: 363-373, 1985.
155. Mykles DL. Crustacean muscle plasticity: Molecular mechanisms determining mass and contractile properties. *Comp Biochem Physiol B Biochem Mol Biol* 117: 367-378, 1997.
156. Naples SP, Borengasser SJ, Rector RS, Uptergrove GM, Morris EM, Mikus CR, Koch LG, Britton SL, Ibdah JA, Thyfault JP. Skeletal muscle mitochondrial and metabolic responses to a high-fat diet in female rats bred for high and low aerobic capacity. *Appl Physiol Nutr Metab* 35: 151-162, 2010.
157. Naya FJ, Mercer B, Shelton J, Richardson JA, Williams RS, Olson EN. Stimulation of slow skeletal muscle fiber gene expression by calcineurin in vivo. *J Biol Chem* 275: 4545-8, 2000.
158. Nishikawa KC, Monroy JA, Uyeno T, Yeo SH, Pai D, Lindstedt S. Is titin a "winding filament"? A new twist on muscle contraction. *Proc Royal Soc* 279: 981-990, 2012.
159. Niven JE, Scharlemann JP. Do insect metabolic rates at rest and during flight scale with body mass? *Biol Lett* 1: 346-9, 2005.
160. Nudds RL, Taylor GK, Thomas AL. Tuning of Strouhal number for high propulsive efficiency accurately predicts how wingbeat frequency and stroke amplitude relate and scale with size and flight speed in birds. *Proc Biol Sci* 271: 2071-2076, 2004.

161. O'Connell B, Nguyen LT, Stephenson GM. A single-fibre study of the relationship between MHC and TnC isoform composition in rat skeletal muscle. *Biochem J* 378: 269-274, 2004.
162. Oakley CR, Gollnick PD. Conversion of rat muscle fiber types. A time course study. *Histochemistry* 83: 555-560, 1985.
163. Okamoto K, Shaw JM. Mitochondrial morphology and dynamics in yeast and multicellular eukaryotes. *Annu Rev Genet* 39: 503-536, 2005.
164. Ono S. Dynamic regulation of sarcomeric actin filaments in striated muscle. *Cytoskeleton (Hoboken)* 67: 677-692, 2010.
165. Pandorf CE, Haddad F, Roy RR, Qin AX, Edgerton VR, Baldwin KM. Dynamics of myosin heavy chain gene regulation in slow skeletal muscle: Role of natural antisense RNA. *J Biol Chem* 281: 38330-38342, 2006.
166. Parsons SA, Millay DP, Wilkins BJ, Bueno OF, Tsika GL, Neilson JR, Liberatore CM, Yutzey KE, Crabtree GR, Tsika RW, Molkentin JD. Genetic loss of calcineurin blocks mechanical overload-induced skeletal muscle fiber type switching but not hypertrophy. *J Biol Chem* 279: 26192-26200, 2004.
167. Parsons SA, Wilkins BJ, Bueno OF, Molkentin JD. Altered skeletal muscle phenotypes in calcineurin A α and A β gene-targeted mice. *Mol Cell Biol* 23: 4331-4343, 2003.
168. Patch LD, Brooks GA. Effects of training on VO₂ max and VO₂ during two running intensities in rats. *Pflügers Arch* 386: 215-219, 1980.
169. Pennycuik C. Wingbeat frequency of birds in steady cruising flight: New data and improved predictions. *J Exp Biol* 199: 1613-1618, 1996.
170. Pennycuik CJ. *Newton Rules Biology: A Physical Approach to Biological Problems*. Oxford: New York: Oxford University Press, 1992.
171. Periasamy M, Kalyanasundaram A. SERCA pump isoforms: Their role in calcium transport and disease. *Muscle Nerve* 35: 430-442, 2007.
172. Pette D, Spamer C. Metabolic properties of muscle fibers. *Fed Proc* 45: 2910-2914, 1986.
173. Piazzesi G, Reconditi M, Linari M, Lucii L, Bianco P, Brunello E, Decostre V, Stewart A, Gore DB, Irving TC, Irving M, Lombardi V. Skeletal muscle performance determined by modulation of number of myosin motors rather than motor force or stroke size. *Cell* 131: 784-795, 2007.
174. Piersma T. Phenotypic flexibility during migration: Optimization of organ size contingent on the risks and rewards of fueling and flight? *J Avian Biol* 29: 511-520, 1998.
175. Piersma T. Why marathon migrants get away with high metabolic ceilings: Towards an ecology of physiological restraint. *J Exp Biol* 214: 295-302, 2011.
176. Postel U, Thompson F, Barker G, Viney M, Morris S. Migration-related changes in gene expression in leg muscle of the Christmas Island red crab *Gecarcoidea natalis*: Seasonal preparation for long-distance walking. *J Exp Biol* 213: 1740-1750, 2010.
177. Rall JA. Energetic aspects of skeletal muscle contraction: Implications of fiber types. *Exerc Sport Sci Rev* 13: 33-74, 1985.
178. Rao A, Luo C, Hogan PG. Transcription factors of the NFAT family: Regulation and function. *Annu Rev Immunol* 15: 707-747, 1997.
179. Reich TE, Lindstedt SL, LaStayo PC, Pierotti DJ. Is the spring quality of muscle plastic? *Am J Physiol Regul Integr Comp Physiol* 278: R1661-R1666, 2000.
180. Reichmann H, Hoppeler H, Mathieu-Costello O, von Bergen F, Pette D. Biochemical and ultrastructural changes of skeletal muscle mitochondria after chronic electrical stimulation in rabbits. *Pflügers Arch* 404: 1-9, 1985.
181. Rinaldi C, Haddad F, Bodell PW, Qin AX, Jiang W, Baldwin KM. Intergenic bidirectional promoter and cooperative regulation of the IIX and IIB MHC genes in fast skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 295: R208-R218, 2008.
182. Roberts TJ. The integrated function of muscles and tendons during locomotion. *Comp Biochem Physiol A Mol Integr Physiol* 133: 1087-1099, 2002.
183. Roberts TJ, Marsh RL. Probing the limits to muscle-powered accelerations: Lessons from jumping bullfrogs. *J Exp Biol* 206: 2567-2580, 2003.
184. Roberts TJ, Weber JM, Hoppeler H, Weibel ER, Taylor CR. Design of the oxygen and substrate pathways. II. Defining the upper limits of carbohydrate and fat oxidation. *J Exp Biol* 199: 1651-8, 1996.
185. Rome LC. Design and function of superfast muscles: New insights into the physiology of skeletal muscle. *Annu Rev Physiol* 68: 193-221, 2006.
186. Rome LC, Klimov AA. Superfast contractions without superfast energetics: ATP usage by SR-Ca²⁺-pumps and crossbridges in toadfish swimbladder muscle. *J Physiol* 526 (Pt 2): 279-286, 2000.
187. Rome LC, Lindstedt SL. The quest for speed: Muscles built for high-frequency contractions. *News Physiol Sci* 13: 261-268, 1998.
188. Rome LC, Syme DA, Hollingworth S, Lindstedt SL, Baylor SM. The whistle and the rattle: The design of sound producing muscles. *Proc Natl Acad Sci U S A* 93: 8095-100, 1996.
189. Rowell LB, O'Leary DS, Kellogg Jr. DL. Integration of cardiovascular control systems in dynamic exercise. *Handbook of Physiology*. Section 12. Exercise: regulation and integration of multiple systems. eds. LB Rowell, JT Shepherd. pp. 770-838, 1996.
190. Roy RR, Pierotti DJ, Garfinkel A, Zhong H, Baldwin KM, Edgerton VR. Persistence of motor unit and muscle fiber types in the presence of inactivity. *J Exp Biol* 211: 1041-1049, 2008.
191. Sagan L. On the origin of mitosing cells. *J Theor Biol* 14: 255-274, 1967.
192. Schaeffer P, Conley K, Lindstedt S. Structural correlates of speed and endurance in skeletal muscle: The rattlesnake tailshaker muscle. *J Exp Biol* 199: 351-358, 1996.
193. Schaeffer PJ, Nichols SD, Lindstedt SL. Chronic electrical stimulation drives mitochondrial biogenesis in skeletal muscle of a lizard, *Varanus exanthematicus*. *J Exp Biol* 210: 3356-3360, 2007.
194. Schaeffer PJ, Villarin JJ, Lindstedt SL. Chronic cold exposure increases skeletal muscle oxidative structure and function in *Monodelphis domestica*, a marsupial lacking brown adipose tissue. *Physiol Biochem Zool* 76: 877-887, 2003.
195. Schaeffer PJ, Wende AR, Magee CJ, Neilson JR, Leone TC, Chen F, Kelly DP. Calcineurin and calcium/calmodulin-dependent protein kinase activate distinct metabolic gene regulatory programs in cardiac muscle. *J Biol Chem* 279: 39593-39603, 2004.
196. Schiaffino S, Reggiani C. Molecular diversity of myofibrillar proteins: Gene regulation and functional significance. *Physiol Rev* 76: 371-423, 1996.
197. Schilling N, Deban SM. Fiber-type distribution of the perivertebral musculature in *Ambystoma*. *J Morphol* 271: 200-214, 2010.
198. Schmidt-Nielsen K. Locomotion: Energy cost of swimming, flying, and running. *Science* 177: 222-8, 1972.
199. Schmidt-Nielsen K. *Scaling: Why is Animal Size So Important?* Cambridge: Cambridge University Press, 1984.
200. Seipel K, Schmid V. Evolution of striated muscle: Jellyfish and the origin of triploblasty. *Dev Biol* 282: 14-26, 2005.
201. Shadwick RE, Syme DA. Thunniform swimming: Muscle dynamics and mechanical power production of aerobic fibres in yellowfin tuna (*Thunnus albacares*). *J Exp Biol* 211: 1603-1611, 2008.
202. Sharp NC. Timed running speed of a cheetah (*Acinonyx jubatus*). *J Zool* 241: 493-494, 1997.
203. Skoglund CR. Functional analysis of swim-bladder muscles engaged in sound production of the toadfish. *J Biophys Biochem Cytol* 10: 187-200, 1961.
204. Staron RS. Human skeletal muscle fiber types: Delineation, development, and distribution. *Can J Appl Physiol* 22: 307-327, 1997.
205. Steinberg GR, Kemp BE. AMPK in health and disease. *Physiol Rev* 89: 1025-1078, 2009.
206. Steudel K. The work and energetic cost of locomotion. II. Partitioning the cost of internal and external work within a species. *J Exp Biol* 154: 287-303, 1990.
207. Stienen GJ, Kiers JL, Bottinelli R, Reggiani C. Myofibrillar ATPase activity in skinned human skeletal muscle fibres: Fibre type and temperature dependence. *J Physiol* 493 (Pt 2): 299-307, 1996.
208. Stockdale FE. Mechanisms of formation of muscle fiber types. *Cell Struct Funct* 22: 37-43, 1997.
209. Suarez RK, Lighton JR, Brown GS, Mathieu-Costello O. Mitochondrial respiration in hummingbird flight muscles. *Proc Natl Acad Sci U S A* 88: 4870-3, 1991.
210. Suarez RK, Staples JF, Lighton JR, Mathieu-Costello O. Mitochondrial function in flying honeybees (*Apis mellifera*): Respiratory chain enzymes and electron flow from complex III to oxygen. *J Exp Biol* 203: 905-911, 2000.
211. Sutko JL, Airey JA. Ryanodine receptor Ca²⁺ release channels: Does diversity in form equal diversity in function? *Physiol Rev* 76: 1027-1071, 1996.
212. Syme DA, Shadwick RE. Effects of longitudinal body position and swimming speed on mechanical power of deep red muscle from skip-jack tuna (*Katsuwonus pelamis*). *J Exp Biol* 205: 189-200, 2002.
213. Syrovoy I, Gutmann E. Myosin from fast and slow skeletal and cardiac muscles of mammals of different size. *Physiol Bohemoslov* 24: 325-334, 1975.
214. Tada M, Kirchberger MA, Repke DI, Katz AM. The stimulation of calcium transport in cardiac sarcoplasmic reticulum by adenosine 3':5'-monophosphate-dependent protein kinase. *J Biol Chem* 249: 6174-6180, 1974.
215. Taylor CR. Relating mechanics and energetics during exercise. *Adv Vet Sci Comp Med* 38A: 181-215, 1994.
216. Taylor CR, Heglund NC. Energetics and mechanics of terrestrial locomotion. *Annu Rev Physiol* 44: 97-107, 1982.
217. Taylor CR, Longworth KE, Hoppeler H. Matching O₂ delivery to O₂ demand in muscle: II. Allometric variation in energy demand. *Adv Exp Med Biol* 227: 171-181, 1988.
218. Taylor CR, Maloiy GM, Weibel ER, Langman VA, Kamau JM, Seeherman HJ, Heglund NC. Design of the mammalian respiratory system. III Scaling maximum aerobic capacity to body mass: Wild and domestic mammals. *Respir Physiol* 44: 25-37, 1981.

219. Taylor CR, Schmidt-Nielsen K, Raab JL. Scaling of energetic cost of running to body size in mammals. *Am J Physiol* 219: 1104-7, 1970.
220. Taylor CR, Weibel ER. Design of the mammalian respiratory system. I. Problem and strategy. *Respir Physiol* 44: 1-10, 1981.
221. Taylor CR, Weibel ER, Weber JM, Vock R, Hoppeler H, Roberts TJ, Brichon G. Design of the oxygen and substrate pathways. I. Model and strategy to test symmorphosis in a network structure. *J Exp Biol* 199: 1643-9, 1996.
222. Taylor GK, Nudds RL, Thomas AL. Flying and swimming animals cruise at a Strouhal number tuned for high power efficiency. *Nature* 425: 707-711, 2003.
223. Torrella JR, Fouces V, Palomeque J, Viscor G. Comparative skeletal muscle fibre morphometry among wild birds with different locomotor behaviour. *J Anat* 192 (Pt 2): 211-222, 1998.
224. van Ekeren GJ, Sengers RC, Stadhouders AM. Changes in volume densities and distribution of mitochondria in rat skeletal muscle after chronic hypoxia. *Int J Exp Pathol* 73: 51-60, 1992.
225. Vock R, Hoppeler H, Claassen H, Wu DX, Billeter R, Weber JM, Taylor CR, Weibel ER. Design of the oxygen and substrate pathways. VI. structural basis of intracellular substrate supply to mitochondria in muscle cells. *J Exp Biol* 199: 1689-1697, 1996.
226. Wakeling JM, Blake OM, Wong I, Rana M, Lee SS. Movement mechanics as a determinate of muscle structure, recruitment and coordination. *Philos Trans R Soc Lond B Biol Sci* 366: 1554-1564, 2011.
227. Wakeling JM, Johnston IA. Muscle power output limits fast-start performance in fish. *J Exp Biol* 201: 1505-1526, 1998.
228. Wang K, McClure J, Tu A. Titin: Major myofibrillar components of striated muscle. *Proc Natl Acad Sci U S A* 76: 3698-702, 1979.
229. Wang LC, Kernell D. Fibre type regionalisation in lower hindlimb muscles of rabbit, rat and mouse: A comparative study. *J Anat* 199: 631-643, 2001.
230. Webb PW. The swimming energetics of trout. I. Thrust and power output at cruising speeds. *J Exp Biol* 55: 489-520, 1971.
231. Weber JM. Metabolic fuels: Regulating fluxes to select mix. *J Exp Biol* 214: 286-294, 2011.
232. Weber JM, Brichon G, Zwingerstein G, McClelland G, Saucedo C, Weibel ER, Taylor CR. Design of the oxygen and substrate pathways. IV. Partitioning energy provision from fatty acids. *J Exp Biol* 199: 1667-1674, 1996.
233. Weber JM, Roberts TJ, Vock R, Weibel ER, Taylor CR. Design of the oxygen and substrate pathways. III. Partitioning energy provision from carbohydrates. *J Exp Biol* 199: 1659-1666, 1996.
234. Weibel ER. *The Pathway for Oxygen : Structure and Function in the Mammalian Respiratory System*. London: Harvard University Press, Cambridge, Massachusetts, 1984.
235. Weibel ER. Scaling of structural and functional variables in the respiratory system. *Annu Rev Physiol* 49: 147-159, 1987.
236. Weibel ER. *Symmorphosis: On form and Function in Shaping Life*. Cambridge, Massachusetts: Harvard University Press, 2000.
237. Weibel ER, Bacigalupe LD, Schmitt B, Hoppeler H. Allometric scaling of maximal metabolic rate in mammals: Muscle aerobic capacity as determinant factor. *Respir Physiol Neurobiol* 140: 115-132, 2004.
238. Weibel ER, Taylor CR, Hoppeler H. The concept of symmorphosis: A testable hypothesis of structure-function relationship. *Proc Natl Acad Sci U S A* 88: 10357-10361, 1991.
239. Weibel ER, Taylor CR, Weber JM, Vock R, Roberts TJ, Hoppeler H. Design of the oxygen and substrate pathways. VII. Different structural limits for oxygen and substrate supply to muscle mitochondria. *J Exp Biol* 199: 1699-1709, 1996.
240. Wells DJ. Muscle performance in hovering hummingbirds. *J Exp Biol* 178: 39-57, 1993.
241. Wickler SJ, Hoyt DF, Clayton HM, Mullineaux DR, Cogger EA, Sandoval E, McGuire R, Lopez C. Energetic and kinematic consequences of weighting the distal limb. *Equine Vet J* 36: 772-777, 2004.
242. Wigmore PM, Evans DJ. Molecular and cellular mechanisms involved in the generation of fiber diversity during myogenesis. *Int Rev Cytol* 216: 175-232, 2002.
243. Williams P, Watt P, Bicic V, Goldspink G. Effect of stretch combined with electrical stimulation on the type of sarcomeres produced at the ends of muscle fibers. *Exp Neurol* 93: 500-509, 1986.
244. Williams TM, Dobson GP, Mathieu-Costello O, Morsbach D, Worley MB, Phillips JA. Skeletal muscle histology and biochemistry of an elite sprinter, the African cheetah. *J Comp Physiol B* 167: 527-535, 1997.
245. Wilson RS, James RS. Constraints on muscular performance: Trade-offs between power output and fatigue resistance. *Proc Soc B* 271: S222-S225, 2004.
246. Woledge RC. The energetics of tortoise muscle. *J Physiol* 197: 685-707, 1968.
247. Wu H, Olson EN. Activation of the MEF2 transcription factor in skeletal muscles from myotonic mice. *J Clin Invest* 109: 1327-1333, 2002.
248. Wu H, Rothermel B, Kanatous S, Rosenberg P, Naya FJ, Shelton JM, Hutcheson KA, DiMaio JM, Olson EN, Bassel-Duby R, Williams RS. Activation of MEF2 by muscle activity is mediated through a calcineurin-dependent pathway. *EMBO J* 20: 6414-6423, 2001.
249. Yales FE. Comparative physiology: Compared to what? *Am J Physiol* 237: R1-R2, 1979.
250. Yeh LA, Lee KH, Kim KH. Regulation of rat liver acetyl-CoA carboxylase. Regulation of phosphorylation and inactivation of acetyl-CoA carboxylase by the adenylate energy charge. *J Biol Chem* 255: 2308-2314, 1980.
251. Zhang MY, Zhang WJ, Medler S. The continuum of hybrid IIX/IIB fibers in normal mouse muscles: MHC isoform proportions and spatial distribution within single fibers. *Am J Physiol Regul Integr Comp Physiol* 299: R1582-R1591, 2010.